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Recent advances and future perspectives in the valorization of olive biomass for sustainable biorefinery applications

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ABSTRACT

This review provides a comprehensive and up-to-date analysis of the valorization pathways of olive-derived biomasses—including pruning residues, leaves, pomace, and stones—as renewable feedstocks for developing sustainable biorefinery systems. Emphasis is placed on recent advances (2019–2024) in thermochemical, biochemical, and emerging conversion routes enabling the production of biofuels, bioenergy, biochemicals, and bio-based materials such as bioethanol, biomethane, hydrogen, bioplastics, and activated carbon. Novel approaches, including deep eutectic solvents and carbon capture integration, are discussed to enhance the process efficiency and sustainability of olive-based biorefineries. In addition to reviewing compositional characteristics and conversion pathways, the study integrates bibliometric and techno-economic insights and proposes a structured SWOT analysis. The review highlights current research gaps and outlines strategic directions to optimize system integration, reduce costs, and environmental impacts. Ultimately, this work positions olive biomass-based biorefineries as a potential pillar towards climate-resilient agro-industrial systems, but unresolved threats/weakness should be solved.

1. Introduction

Growing concerns about climate change, resource depletion, and environmental degradation have intensified the search for alternative solutions to replace fossil-based energy, chemicals, and materials. Within this context, biorefineries—facilities designed to transform biomass into various value-added bioproducts—have emerged as pivotal in transitioning to a more sustainable and circular bioeconomy [1]. Replicating the operational logic of conventional fossil-based refineries, biorefineries utilize renewable biomass feedstocks, potentially mitigating environmental impact, reducing dependence on fossil resources, and valorizing agro-foresty and industrial bio-based residues and by-products.

One such renewable bio-based feedstock is the olive biomass—including pruning residues, leaves, pomace, and stones—representing a promising yet underexploited resource. Predominantly generated in Mediterranean countries, these olive-based biomasses exceed 35 million tons annually [2], but are mostly destined for low-value uses such as combustion or soil amendment. However, their rich lignocellulosic and bioactive composition enables their transformation into bioenergy, biofuels, biochemicals, and advanced materials [2–5], contributing to both waste valorization and climate mitigation. Conversion routes encompass thermochemical, biochemical, and chemical processes leading to the production of bioethanol, biomethane, biodiesel, bioplastics, biochar, and activated carbon, among others [3–7]. Furthermore, phenolic compounds extracted from olive residues are gaining interest due to their antioxidant properties and applications in the pharmaceutical, food, and cosmetic sectors [4]. Integrating olive biomass into biorefinery platforms would help reduce waste and contribute to resource efficiency, GHG emission reduction, and rural development, aligning with the principles of circularity and sustainability [8].

Over the past decade, there has been growing scientific interest in the valorization of olive biomass as a versatile and abundant feedstock. Prior review articles primarily concentrated on its physical characteristics and potential for energy generation, with a particular focus on pruning residues, olive stones, and pomace. These studies explored their applications in the production of bioethanol, biochar, biocomposites, animal feed, activated carbon, and cellulose pulp, as well as their use in

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Abbreviations

ABE	Acetone-Butanol-Ethanol
BECCS	Bioenergy carbon capture and storage
CCS	Carbon capture and storage
CCU	Carbon capture and utilization
DES	Deep eutectic solvents
FPU	Filter paper unit
GHG	Green House Gases
HTC	Hydrothermal carbonization
HMF	Hydroxymethylfurfural
HTR	Hydraulic retention time
LCA	Life cycle assessment
NADES	Natural deep eutectic solvents
OLR	Organic loading rates
PHA	Polyhydroxyalkanoate
RFPT	Radio frequency plasma torch
R&D	Research and development
PSSF	Pre-saccharification and simultaneous saccharification and fermentation
S/L	Solid/liquid
SSF	Simultaneous saccharification and fermentation
SDGs	Sustainable development goals
SWOT	Strengths, Weaknesses, Opportunities, and Threats
TS	Total Solid
VS	Volatile Solids

thermochemical conversion processes such as combustion, gasification, and pyrolysis [9–14]. Parallel to these investigations, the treatment and valorization of olive mill wastewater have also received considerable attention, particularly regarding the recovery of valuable components and mitigation of environmental impacts [15,16]. Other reviews have shifted towards a deeper investigation of thermochemical technologies—especially pyrolysis and gasification—for thermal and energy applications [17–19]. More recently, scholarly attention has expanded to encompass advanced conversion routes for the development of high-value products. These include the synthesis of novel materials [13, 19] and the extraction of bioactive compounds such as phenolics with potential applications in the pharmaceutical and food sectors [20,21].

This review provides a critical, updated, and integrative synthesis of recent advances (2018–2024) in valorizing olive biomass within biorefinery systems, expanding the scope and analytical depth of previous reviews. It begins with a bibliometric analysis to trace the evolution of scientific interest and identify thematic clusters in the field. A detailed characterization of the main olive-derived biomasses—pruning residues, olive leaves, pomace, exhausted pomace, and stones—is then provided, including their availability and chemical composition.

Building on this foundation, the review then examines thermochemical, biochemical, and chemical conversion pathways, highlighting recent technological findings and their application in the production of energy carriers (e.g., bioethanol, biogas, hydrogen), high-value chemicals (e.g., polyphenols, organic acids, xylitol), and advanced materials (e.g., biochar, biopolymers, carbon dots). Quantitative and qualitative assessments are integrated to compare performance, environmental implications, and feasibility across different routes. Finally, a SWOT analysis is employed to identify strengths, weaknesses, opportunities, and threats associated with the implementation of olive-based biorefineries to conclude by proposing future directions to improve integration, reduce costs, and enhance sustainability. By offering a comprehensive and data-driven perspective, this review positions olive biomass as a strategic feedstock for building climate-resilient and circular bioeconomy systems, with relevance that extends beyond Mediterranean regions.

2. Bibliometric analysis

To establish a robust foundation for this review, a bibliometric analysis was conducted to assess the evolution of scientific interest in olive biomass valorization within biorefinery systems. Fig. 1 presents key insights into the research landscape, highlighting publication trends, disciplinary distribution, and dominant thematic clusters.

The bar chart (Fig. 1a, top-left) illustrates a continuous increase in the number of publications on olive biomass valorization from 2019 to 2024 with journal articles constituting the predominant publication type. This trend reflects a growing academic commitment to advancing sustainable biorefinery solutions for olive-derived residues.

The treemap (Fig. 1b, top-right) provides insights into the disciplinary composition of the field focused on biorefinery and olive biomass. The largest contributions come from Environmental Science (17.9 %), followed by Energy (13.6 %) and Agricultural and Biological Sciences (13.3 %), underscoring the sustainability-driven nature of this research. The presence of Chemical Engineering (9.9 %), Biochemistry, Genetics, and Molecular Biology (7.8 %), and Materials Science (3.2 %) reflects the increasing focus on advanced valorization pathways such as biochemical extraction, biofuels and platform chemicals production, bio-based material development, and thermochemical conversion techniques. Among these categories, Environmental Science in Scopus—the primary search engine used to retrieve publications in this work—is generally one of the most relevant in the field of bibliometrics [22]. It encompasses a wide range of research areas related to environmental chemistry, environmental engineering, and other relevant multidisciplinary fields related to sustainability, climate change, environmental impact/pollution, health, and social sciences, which can help explain this trend.

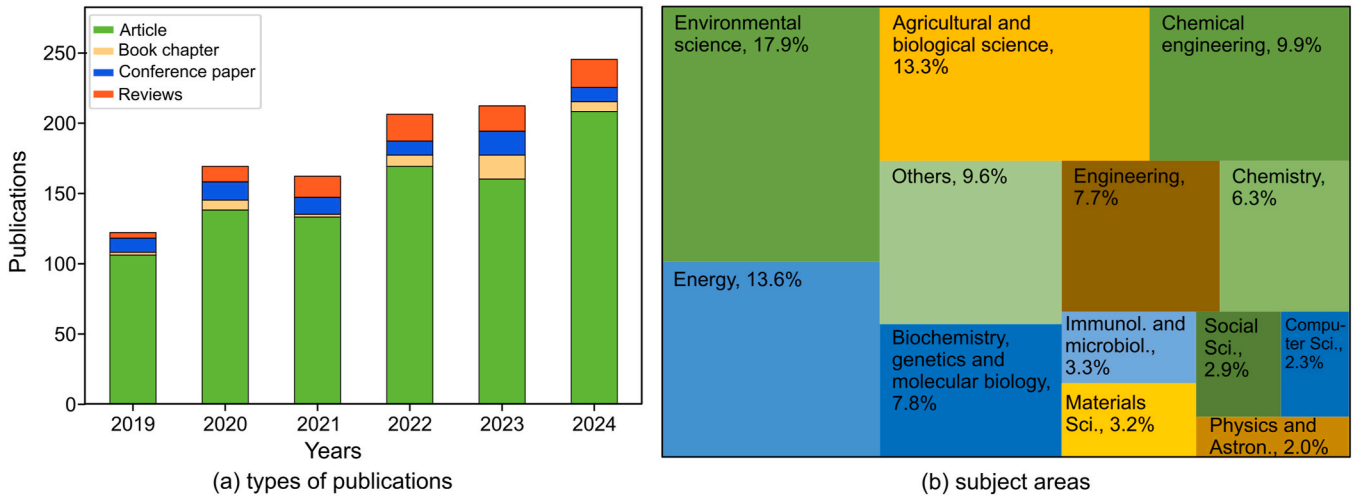
The VOSviewer keyword network (Fig. 1c, bottom) visualizes key thematic clusters within the field, with major nodes representing highly interconnected research areas [23]. The green cluster is centered on biomass, olive pomace, and thermochemical conversion, reflecting an interest in primarily energy recovery and, secondly, biochar production. The blue cluster focuses on olive mill wastewaters, anaerobic digestion, and biogas, highlighting biochemical valorization strategies to produce energy products along with other bioproducts. The purple cluster, led by circular economy and sustainability, underscores the systemic relevance of olive biomass valorization for climate-resilient bioeconomy models. The red cluster is associated with polyphenols and bioactive compounds, showing the increasing importance of olive biomass in producing high-value biochemicals. Overall, the bibliometric findings confirm the interdisciplinary nature of olive biomass valorization, spanning technological, environmental, and economic aspects. The growing research output underscores its strategic role in sustainable agro-industrial systems. Building on these insights, this review integrates recent advancements, identifies research gaps, and explores pathways to enhance the efficiency and scalability of olive biomass biorefineries.

3. Availability and characterization of olive-based biomass for its valorization

3.1. Olive oil production and availability of olive-based biomass

Given the wide scope of olive oil production and biomass generation, we present a concise overview of key biomass fractions, their composition, potential uses, and related quantitative data. For further insights into production dynamics and biomass availability, refer to specialized sources.

The global production of edible oils continues to grow, with olive oil



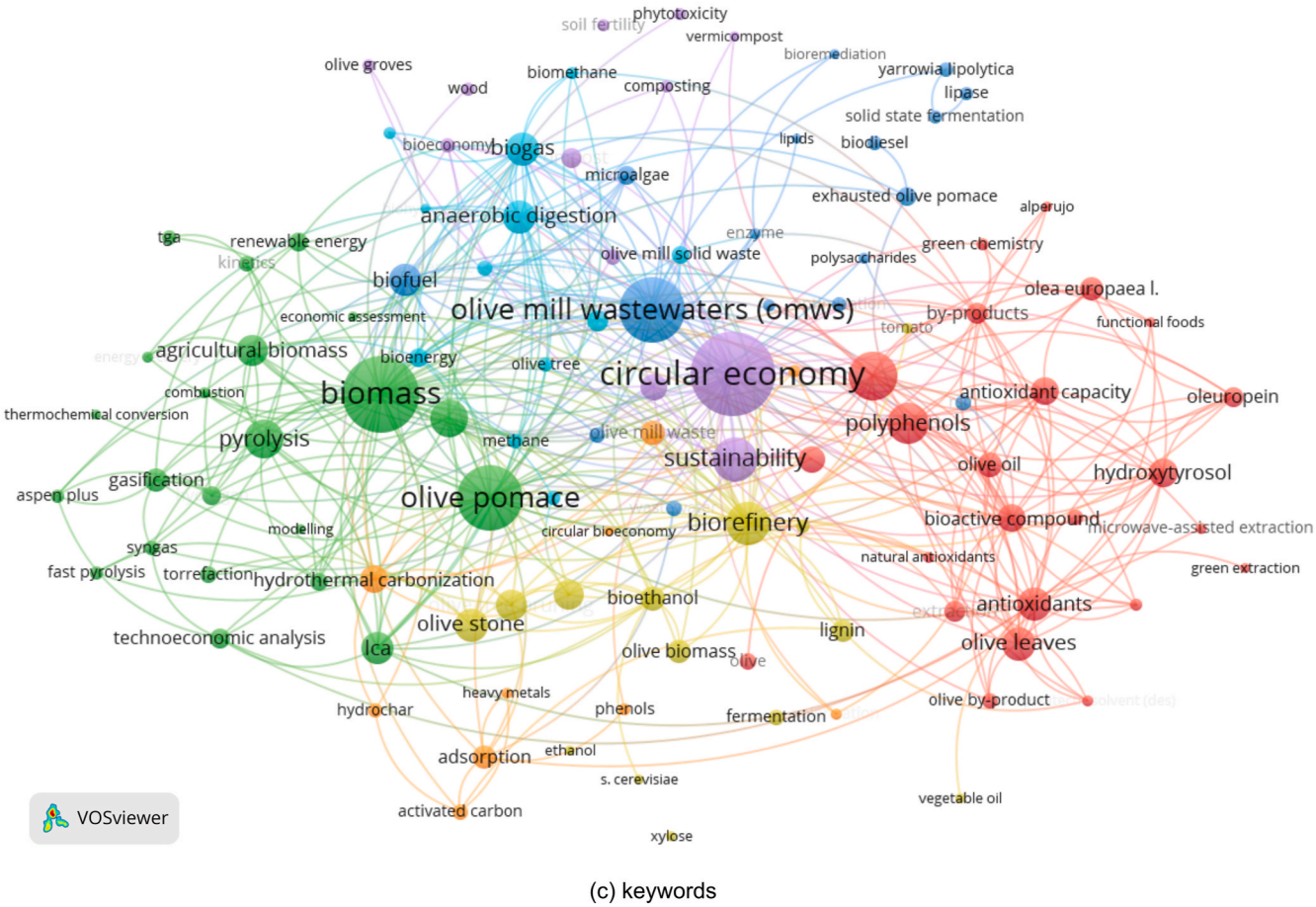


Fig. 1. Publication trends in olive biomass valorization since 2019: a) Number of publications by type and year, b) subject areas of the publications, and c) keyword co-occurrence analysis carried out with VOSviewer. Abbreviations in subplot b: Astron., astronomy; immunol., immunology; microbiol., microbiology; sci., sciences.

accounting for 1.6 % of total production in 2023, following an increasing consumption trend driven by its recognized health benefits [2,24]. Despite fluctuations caused by climate and market conditions, the Mediterranean region remains the dominant hub, contributing over 70 % of global olive oil production, with Spain, Italy, and Greece as leading producers (Fig. 2) [25,26].

Olive cultivation spans over 40 countries, yet leading countries—Spain, Italy, Tunisia, Turkey, Greece, Portugal, and Morocco—account for ~70 % of global olive oil output. The key production areas are Andalusia (Spain), Apulia (Italy), and the Peloponnese (Greece) (Fig. 2). Fig. 3 shows the fluctuations in olive oil production and the consumption trend over the last twenty years. The observed fluctuations in production will directly influence biomass availability in these regions, which varies based on climatic conditions, including droughts, extreme weather events, and other environmental challenges, as well as socio-political and economic factors that can disrupt supply chains and market stability. Other factors that can affect olive biomass availability include agronomic practices and extraction processes, which vary between producer countries [4]. Olive oil extraction generates diverse biomass residues, broadly classified into agricultural biomass (olive tree

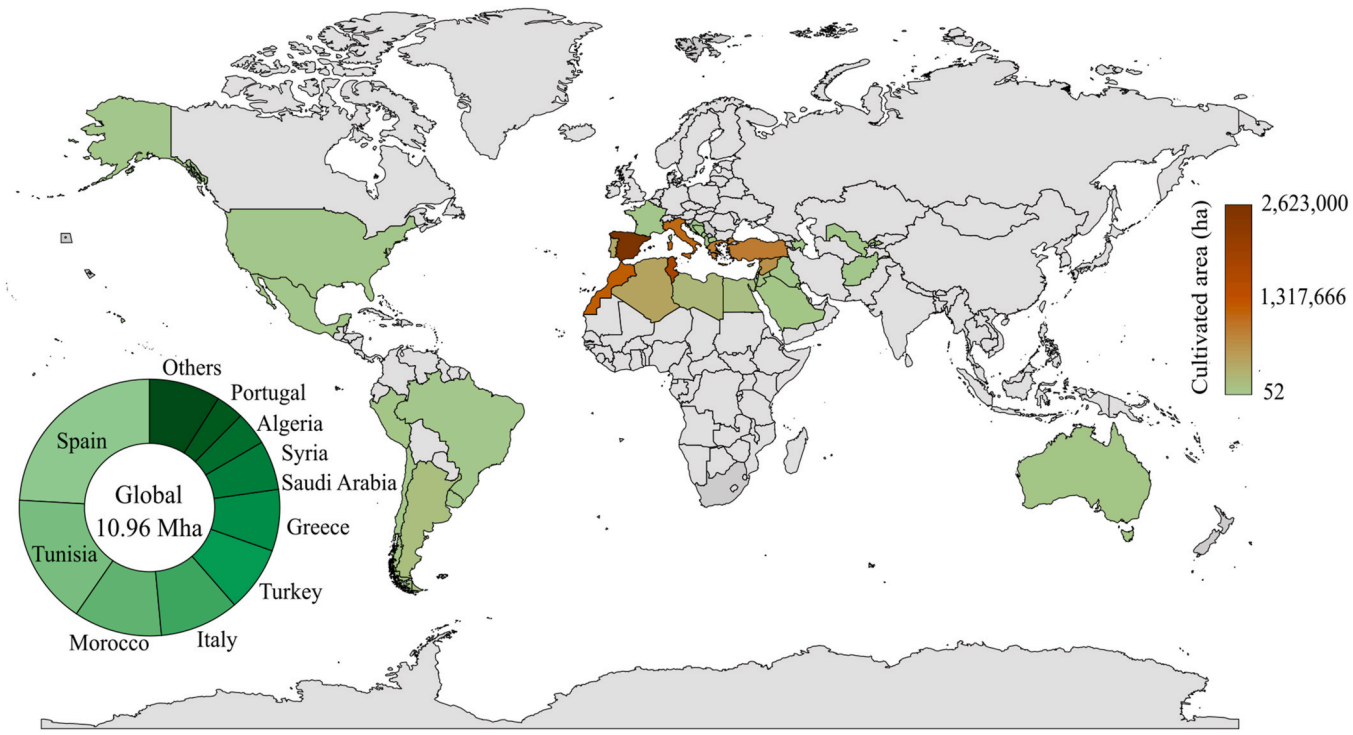


Fig. 2. Global distribution of olive groves (in hectares) by country in 2022.

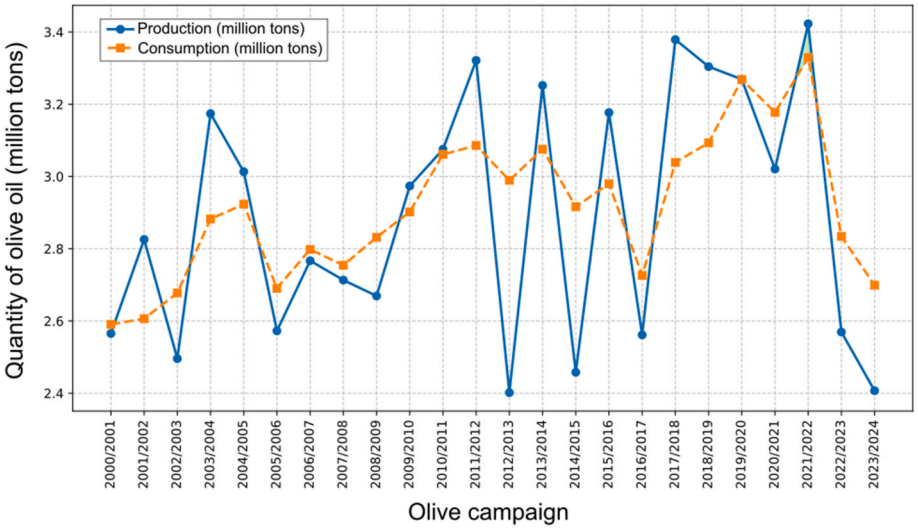


Fig. 3. World production and consumption of olive oil from 2000 to 2024.

pruning residues, leaves) and agro-industrial biomass (olive mill leaves, olive pomace, exhausted pomace, and olive stones). Although traditionally underutilized, these by-products offer strong potential for biorefinery applications, offering sustainable pathways for interest products.

Spain leads globally, with 2.4 million hectares dedicated to olive cultivation. However, recent droughts have caused significant variations in olive oil production (0.67–0.77 million tons in 2022/2023 and 2023/2024). Forecasts for 2024/2025 predict a 48 % increase, reaching 1.3 million tons [25], which will likely result in a proportional rise in biomass generation, particularly olive pomace, exhausted pomace, and olive stones, derived from oil extraction. In addition to agro-industrial residues, agricultural biomass follows seasonal cycles and remains an abundant yet underutilized resource. In Spain and the year 2022, olive tree pruning generated an estimated 3.95 million tons, while olive mill leaves, removed during the cleaning process, contribute ~0.3 million tons [2]. Among processing by-products, olive pomace is the most abundant, reaching ~2.2 million tons. Further oil recovery from pomace produces ~0.39 million tons of exhausted olive pomace [25], while olive stones, widely used as biofuel, add ~0.35 million tons [27]. These figures underscore the vast potential for biomass valorization, focusing only on Spain and emphasizing the potential for efficient biorefinery strategies.

3.2. *Characterization of olive-based biomass*

As shown in Table 1, the chemical composition of these olive-based biomasses varies significantly, particularly in terms of lignin, cellulose,

Table 1
Overview of availability and chemical composition of olive biomasses (%, dry weight) [2,25,27,29–31].

	Olive tree pruning	Olive mill leaves	Olive pomace	Exhausted olive pomace	Olive stones[c]
Composition, %					
Ash	3.9	3.7	6.6	3.4–9.1	0.6
Lignin	17.7	19.3	43.2	21.9–41.6	35.6
Hemicellulose	14.5	4.5	22.3	24.1–11.3	26
Cellulose	21.6	11.8	12.5	11.0–10.1	20.9
Fat	NA[b]	9.9	16.7–20.0[a]	2.5, 4	NA[b]
Protein	3.1	7	0.4	0.3–10.6	0.7
Extractives	28.6	38.8	36.8–40.7[a]	48.8–52.9	6.3

[a] Two-phase extraction.
[b] NA, not available.
[c] Estimated using the calculation base.

hemicellulose, extractives, and ash content—factors that directly influence their suitability for specific conversion routes. For instance, high lignin content in olive stones favors thermochemical processes such as pyrolysis or combustion, while the elevated extractives and cellulose content in pruning residues make them promising for biochemical conversions and bioactive compound extraction. Understanding these compositional differences is essential for selecting the most appropriate valorization strategies within integrated biorefinery frameworks [28].

4. Valorization pathways in biorefineries

The diverse availability and rich chemical composition of olive biomass offers valuable opportunities for its integration into second-generation biorefinery systems. These systems can be classified according to product orientation and conversion pathways. On one hand, energy-oriented biorefineries aim to produce heat, electricity or biofuels (liquid and gaseous). On the other hand, product-oriented biorefineries focus on the extraction of bioactive compounds, platform and specialty chemicals (succinic acid, lactic acid, xylitol, furfural, etc.) or materials [8]. This section reviews the main valorization pathways, with emphasis

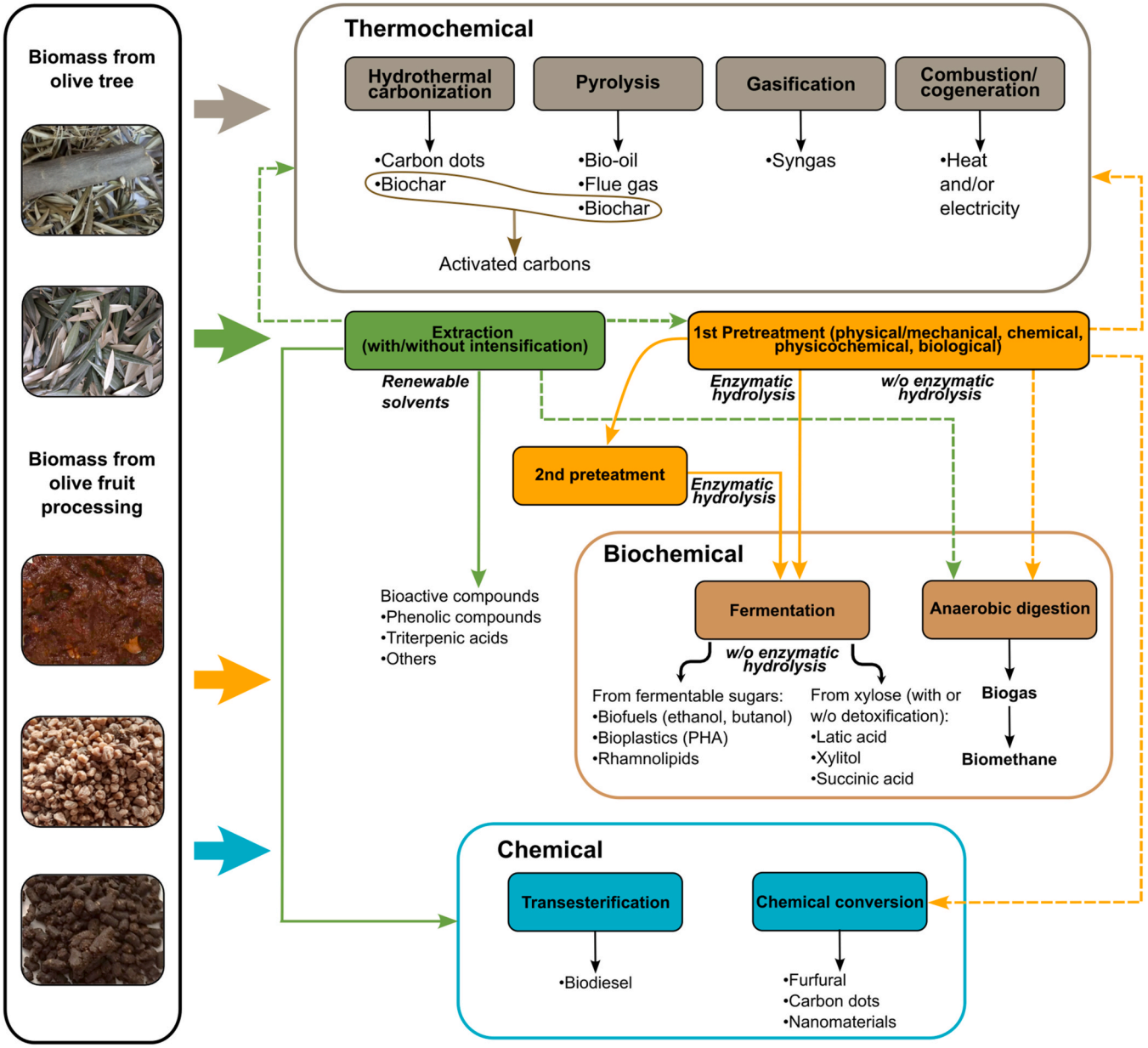


Fig. 4. Valorization pathways for energy and product-driven biorefineries utilizing olive biomass.

on recent developments, technological innovations and potential applications.

As illustrated in Fig. 4, the valorization of olive biomass follows a variety of processing routes depending on the type of feedstock (e.g., leaves, prunings, pomace, stones) and the desired end-products. A first conditioning step often involves mechanical pretreatment, such as milling or pelletizing. However, in many cases, more intensive deconstruction pretreatments are required to disrupt the lignocellulosic matrix and enhance the efficiency of downstream processes. These include treatments with acids, alkalis, steam, or organosolv systems [32], which increase the accessibility of structural components—cellulose, hemicellulose, and lignin—facilitating their conversion into high-value products via biochemical, chemical, or thermochemical routes. Additionally, sequential pretreatment strategies have been investigated to enable multiproduct valorization and to better align the chemical composition of the biomass with the requirements of specific conversion pathways [33].

A particularly relevant line of research focuses on the extraction of bioactive compounds, such as phenolics and triterpenic acids, due to their high added value and applications as antioxidants and antimicrobials. To improve their recovery, advanced extraction techniques—including ultrasound-assisted extraction [34], microwave-assisted extraction [35], and supercritical CO_2 extraction [36]—have been developed. In this context, the use of green and safe solvents is strongly encouraged, with increasing interest in renewable and low-toxicity alternatives such as ethanol, water, natural deep eutectic solvents (NADES) [20], and supercritical CO_2 [4].

To preserve sensitive bioactives and improve the performance of subsequent processes, extraction is typically conducted prior to chemical or biological pretreatment [5,33], reducing potential microbial inhibition in fermentation stages [33].

In terms of conversion technologies, olive biomass can be processed through thermochemical, chemical, or biochemical routes [37]. For wet biomass, hydrothermal methods such as hydrothermal carbonization, liquefaction, and gasification are employed. Hydrothermal carbonization (HTC) (180–250 °C) yields biochar; liquefaction (250–374 °C) produces bio-oil, aqueous organics, light gases, and solid residues; while gasification at >373 °C generates syngas rich in H_2 and CH_4 [38]. For dry biomass, conventional thermochemical processes such as pyrolysis (350–700 °C, in the absence of oxygen), gasification (700–1200 °C, limited oxygen or steam), and combustion (800–1000 °C, with air) are commonly used to produce energy carriers such as bio-oil, biochar, syngas, heat, and electricity [8,39].

Chemical conversion routes involve the transformation of biomass intermediates into valuable products through reactions such as transesterification (e.g., biodiesel production from oils) [40], sugar dehydration to produce furans [41], and cascade reactions for the synthesis of advanced materials like carbon dots [42]. Meanwhile, biochemical pathways are based on the microbial fermentation of carbohydrates and other substrates to generate biofuels such as bioethanol or biobutanol [43,44] and platform chemicals such as succinic acid and lactic acid [45]. Furthermore, anaerobic digestion plays a complementary role, producing biogas and digestate, the latter being suitable as a soil amendment and contributing to nutrient recycling [46]. Altogether, these integrated valorization pathways offer significant opportunities to support the development of regional bioindustries linked to the olive agro-industrial sector. Beyond enabling the diversification of renewable fuels and bioproducts, they play a key role in advancing a low-carbon, circular bioeconomy, as conceptually represented in Fig. 4.

According to Fig. 4, the following sections describe examples of energy products, including liquid fuels (ethanol and butanol) (Section 5.1) and gaseous fuels (biomethane, biohydrogen, and hydrogen from syngas) (Section 5.2), as well as the conversion route and provide a brief overview of the operational conditions. Examples of bio-based chemicals and materials derived from olive biomass are then presented in Sections 6 and 7, respectively, to showcase the broad range of valorization possibilities.

5. Energy products from olive biomass

5.1. Liquid fuels

This section discusses the production of liquid fuels, including bioethanol and butanol, through pretreatment and biochemical conversion. It provides examples of sourced olive biomass and process conditions, depending on whether fermentable sugars are derived from the hemicellulosic and/or cellulose fractions.

The production of bioethanol from lignocellulosic biomass is becoming increasingly important, particularly in the context of second-generation biofuels. Olive-derived biomasses—such as pruning residues, olive mill leaves, olive pomace, and olive stones—are of growing interest due its high polysaccharide content and wide availability. The conventional process includes three main stages, including pretreatment, enzymatic hydrolysis, and fermentation (Fig. 5) [47].

Pretreatment disrupts the lignocellulosic structure, releasing cellulose, hemicellulose, and lignin, and producing two fractions: a solid fraction, enriched in cellulose, and a liquid fraction (pretreatment liquor), containing solubilized hemicellulosic sugars. The solid fraction is typically hydrolyzed enzymatically and then fermented, while the liquid fraction can be fermented directly if detoxified. Hydrolysis converts cellulose to glucose, and fermentation converts these sugars into ethanol. Simultaneous saccharification and fermentation (SSF), sometimes with a short pre-saccharification step (PSSF), is commonly employed to reduce feedback inhibition and operational costs [41]. Fig. 5 illustrates the simplified metabolic pathway of the yeast *Saccharomyces cerevisiae*, which produces two molecules of ethanol from glucose. To further improve ethanol yields, hemicellulosic sugars present in the pretreatment liquor can also be valorized. Since *S. cerevisiae* cannot ferment xylose, non-conventional microorganisms such as *Escherichia coli* SL100 and *Candida guilliermondii* FTI 20037 have been employed [33,48]. Both strains utilize glucose and xylose, with *C. guilliermondii* additionally producing xylitol, thereby enhancing the overall process efficiency in integrated biorefinery schemes.

Several studies have explored bioethanol production from olive-derived biomasses, such as pruning residues, olive mill leaves, pomace, and stones, using various microorganisms and processing strategies (Table 2). A key research focus has been optimizing pretreatment methods, which are essential to break down lignocellulosic structures into fermentable sugars. Techniques such as liquid hot water, acid hydrolysis, steam explosion, microwave-assisted treatment, and their combinations have been investigated. Among these, acid pretreatment has shown promising yields—for example, up to 280 g ethanol/kg sugars from acid-hydrolyzed olive pruning residues fermented with *C. guilliermondii* [48]. However, using sulfuric acid (2–2.5 %) raises environmental concerns that must be carefully considered. Hence, the use of bases in the pretreatment was also applied to olive tree pruning, e.g., calcium hydroxide [49], but it should be considered that detoxification and enzymatic hydrolysis steps were required to ferment the pretreatment liquor. In this case, the production of inhibitors, such as hydroxymethylfurfural (HMF) and phenolic compounds, increased. However, the detoxification process with activated carbon proved efficient, achieving a 33 % increase in efficiency. Improvements in bioethanol production yield from olive stones were also observed with detoxification processes of the pretreatment liquors with *E. coli* SL100 [44]. In this study, the overall yield was 160.1 and 162.2 g ethanol/kg biomass, depending on whether the pretreated solid was also subjected to SSF or not and the microorganisms applied in the fermentation (Table 2). In contrast, olive tree pruning with calcium hydroxide yields 28.4 g/kg biomass [50]. It highlights the importance of the overall conversion route selected, including the enzymes and fermentation microorganisms in improving yields, particularly with Cellic® CTec2 and the use of *E. coli* to ferment xylose-rich hydrolysates from the

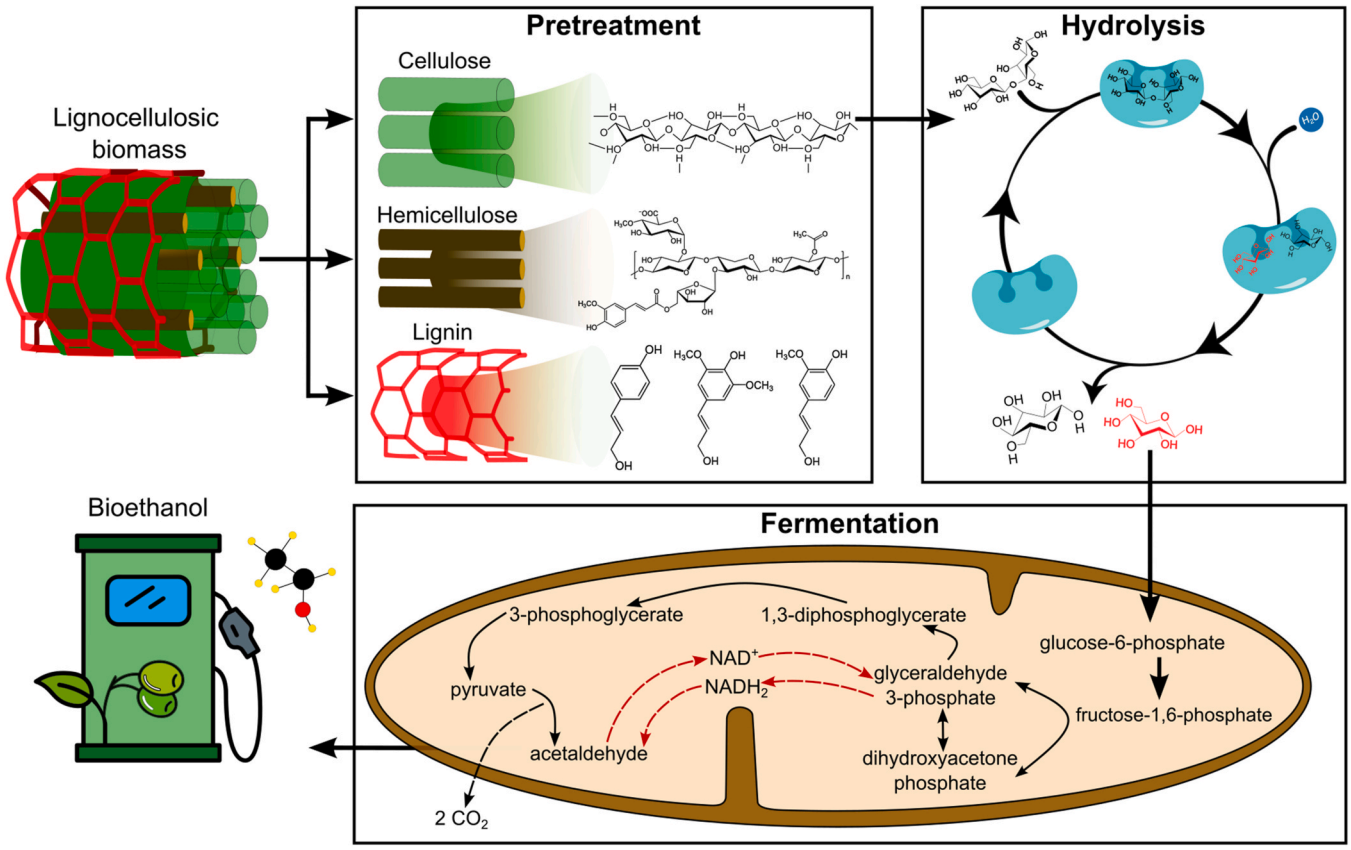


Fig. 5. Diagram of the integrated process for bioethanol production from olive lignocellulosic biomass.

pretreatment and Cellic® CTec2 and *S. cerevisiae* to use cellulose-rich pretreated solids.

New pretreatment technologies are promising such as comparing microwaves with autoclaves for heating. In this case, microwave and formic acid-assisted pretreatments yield 91.5 g of ethanol/kg biomass for olive pomace, but selecting the acid, temperature, and enzyme is crucial [51].

Moreover, the application of two sequential pretreatments has been proposed in some cases to fractionate the olive biomass and obtain bioethanol along with other products (Table 2). For example, the production of bioethanol yielded about 102 g/kg biomass, pretreating olive mill leaves with diluted sulfuric acid (2 % w/v) at 120 °C followed by an organosolv pretreatment catalyzed with sulfuric acid at 150 °C [52]. Using olive stones, an overall value of 162 g bioethanol/kg biomass was achieved using diluted acid pretreatment (10.5 % w/w biomass) at 180 °C and steam explosion at 195 °C [44]. These values were achieved by fermenting both xylose and glucose with *E. coli* and *S. cerevisiae*, respectively.

However, increasing the biomass pretreatment steps does not substantially improve the yield for bioethanol production, and it should be evaluated not only for the yield but also considering techno-economic and environmental aspects. For example, a study applied extraction (recovering phenolic compounds and oligosaccharides) and two sequential pretreatments on the pruning, yielding 80 and 77 g bioethanol/kg biomass when the pretreated solid was fermented with *S. cerevisiae* Ethanol Red®. In this study, the first pretreatment was based on diluted phosphoric acid at 130 °C, and the values depended on the second pretreatment as liquid hot water at 200 °C and steam explosion at 220 °C were applied, respectively [33]. However, another study obtained 103 g of bioethanol/kg biomass by sequentially applying extraction and only one pretreatment by steam explosion catalyzed with phosphoric acid at 195 °C. Moreover, to increase the overall yield to about 180 g bioethanol/kg extracted biomass (~121 g/kg biomass), the latter authors used the xylose-fermenting *S. cerevisiae* F12 to ferment both the liquor and the pretreated solid from the pruning [53].

As an alternative to ethanol, butanol can be produced on an industrial scale by anaerobic bacteria of the genus *Clostridium*, which produces the mixture Acetone-Butanol-Ethanol (ABE) [54]. Previous studies focused on olive mill wastewater to produce butanol using ABE fermentation with *Clostridium* spp [55]. and with anaerobic microflora from this type of residue [43]. However, despite the interest in the production of butanol for fuel purposes, to our knowledge, in the last six years, butanol has not been a targeted bioproduct using olive biomass. The production of inhibitory substances during the pretreatment and hydrolysis stages is one of the main obstacles; these substances can impede fermentation and drastically reduce butanol yields. Consequently, butanol production from olive biomass remains expensive and inefficient, which hinders its large-scale use [56]. To improve the economic viability of butanol production by fermentation, it is necessary to overcome these limitations by developing pretreatment techniques, microbial tolerance, and process optimization.

5.2. *Gaseous fuels*

This section covers the production of gaseous fuels (biogas and hydrogen) from olive biomass and provides examples for their production. Biogas is primarily produced through a biochemical process that operates under mesophilic conditions. For hydrogen, the section discusses both the biochemical process based on fermentation with the *Clostridium* family and thermochemical methods that produce gases rich in hydrogen.

5.2.1. *Biogas and methane*

The biochemical and thermochemical routes can achieve gas fuels.

Table 2
Recent studies in bioethanol production to olive biomass: experimental conditions, selected fractions for fermentation, ethanol concentration, and ethanol yield. When various pretreatments are applied, each one is numbered.

Pretreatment	Hydrolysis and fermentation/Fermentation	Ethanol produced	Ref.
Olive tree pruning			
1. Aqueous extraction: S/L 20 % (w/v) at 120 °C, 60 min 2. Acid pretreatment: H_3PO_4 2.5 % (w/v), S/L 20 % (w/v) at 130 °C, 90 min 3. Steam explosion: 400 g dry solid at 220 °C, 5 min	**Solid fraction**[b]**:** SSF: Enzymes Cellic CTec2 (15 FPU/g) and β-glucosidase (15 IU/g) and fermentation with *S. cerevisiae*[a] 4 % (v/v), S/L 25 % (w/v) at 37 °C, 78 h	35.2 g/L 80.0 g/kg extracted biomass	[33]
1. Aqueous extraction: S/L 20 % (w/v) at 120 °C, 60 min 2. Acid pretreatment: H_3PO_4 2.5 % (w/v), S/L 20 % (w/v) at 130 °C, 90 min 3. Liquid hot water: S/L 20 % (w/v), 200 °C, 5 min	**Solid fraction**[b]**:** SSF: Enzymes Cellic CTec2 (15 FPU/g) and β-glucosidase (15 IU/g) and fermentation with *S. cerevisiae*[a] 4 % (v/v), S/L 25 % (w/v) at 37 °C, 78 h	34.0 g/L 77.0 g/kg extracted biomass	[33]
H_2SO_4 2.5 %, S/L 1:10 at 100 °C, 150 min	**Liquid fraction:** Fermentation: *C. guilliermondii*[a] FTI 20037, 350 rpm, 30 °C	280.0 g/kg sugars	[48]
$Ca(OH)_2$ 0.15 % (w/v) at 160 °C, 120 min	**Liquid fraction:** Hydrolysis: Cellulase from *T. reesei*[a] and hemicellulase from *A. niger*[a] at S/L 40 g/L, 100 rpm, 45 °C, 24 h Detoxification: Activated carbon, 200 mg/100 mL, 25 °C, 60 min Fermentation: *S. cerevisiae*[a] Thermosacc at 100 rpm, 33 °C, 72 h	2.4 g/L 510.0 g/kg sugars	[49]
$Ca(OH)_2$ 1 % (w/w), S/L 5 % (w/v) at 160 °C, 120 min	**Liquid fraction:** Hydrolysis: Cellulase from *T. reesei*[a] and hemicellulase from *A. niger*[a] at S/L 40 g/L, 100 rpm, 45 °C, 24 h Fermentation: *S. cerevisiae*[a] BSc114 at 28 °C, 72 h	28.4 g/kg biomass	[50]
1. Aqueous extraction: S/L 10 % (w/v), 120 °C, 60 min 2. Stream explosion: H_3PO_4 1 % (w/w), 300 g at 195 °C, 10 min	**Liquid fraction:** Detoxification: Microionex MB 200 resin at 0.3 g/L Fermentation: *S. cerevisiae*[a] F12, 35 °C **Solid fraction:** Hydrolysis: Cellic CTec2 (15 FPU/g), 25 % (w/w), 50 °C Fermentation: *S. cerevisiae*[a], 35 °C, 48 h	180.0 g/kg extracted biomass	[53]
NaOH 4 % (w/v), S/L 100 g/L at 100 °C, 60 min	**Solid fraction:** SSF: Cellulase from *A. niger*[a] (30 FPU/g) and fermentation with *P. kudriavzevii*[a] YSR7 at 150 rpm, 42 °C, 72 h	6.3 g/L	[57]
$C_2H_2O_4$ 75 mM, S/L 1:10 at 150 °C, 30 min	**Liquid fraction:** Detoxification conditions: $NaBH_4$, 0.03 mol/L, 30 min Fermentation: *S. stipitis*[a] CBS 6054, 28 °C, 63 h	3.8 g/L 271.0 g/kg substrate	[58]
Olive mill leaves			
1. Acid pretreatment: H_2SO_4 2 % (w/v), S/L 15 % (w/w) at 120 °C, 60 min 2. Organosolv pretreatment: Ethanol 50 % (v/v), H_2SO_4 1 % (w/v), S/L 10 % (w/v) at 150 °C, 60 min	**Liquid fraction:** Fermentation: *E. coli*[a] SL100, 300 rpm, 37 °C, 96 h **Solid fraction:** Hydrolysis: Cellic CTec2 (15 FPU/g) Fermentation: *S. cerevisiae*[a] (Ethanol red), 5 % (w/v), 150 rpm, 37 °C, 72 h	102.0 g/kg biomass	[52]
Olive pomace			
CH_2O_2, 0.6M at 140 °C (microwaves)	**Solid fraction:** Hydrolysis: Cellic CTec2 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisiae*[a] (Ethanol red), 120 rpm, 35 °C, 120 h	15.9 g/L 91.5 g/kg biomass	[51]
H_2O at 140 °C (microwaves)	**Solid fraction:** Hydrolysis: Cellic CTec2 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisiae*[a] (Ethanol red), 120 rpm, 35 °C, 24 h	9.6 g/L 55.1 g/kg biomass	[51]
H_2SO_4, 2 % (w/v) at 140 °C (microwaves)	**Solid fraction:** Hydrolysis: Cellic CTec2 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisae*[a] (Ethanol red), 120 rpm, 35 °C, 24 h	3.5 g/L 19.8 g/kg biomass	[51]
Formic acid 0.6 M at 140 °C (microwaves)	**Solid fraction:** Hydrolysis: Accellerase 1500 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisiae*[a] (Ethanol red), 120 rpm, 35 °C, 48 h	11.6 g/L 67.1 g/kg biomass	[51]
CH_2O_2, 0.6 M at 121 °C (autoclave)	**Solid fraction:** Hydrolysis: Cellic CTec2 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisiae*[a] (Ethanol red), 120 rpm, 35 °C, 24 h	5.6 g/L 32.0 g/kg biomass	[51]
H_2O at 121 °C (autoclave)	**Solid fraction:** Hydrolysis: Cellic CTec2 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisiae*[a] (Ethanol red), 120 rpm, 35 °C, 48 h	0.6 g/L 3.6 g/kg biomass	[51]
H_2SO_4, 2 % (w/v) at 121 °C (autoclave)	**Solid fraction:** Hydrolysis: Cellic CTec2 16 % (w/v), 190 rpm, 50 °C, 24 h Fermentation: *S. cerevisiae*[a] (Ethanol red), 120 rpm, 35 °C, 24 h	2.7 g/L 15.3 g/kg biomass	[51]
Na(OH), 2 % (w/v), 130 °C, 60 min (autoclave)	**Liquid fraction:** Enzyme: Filtrase NLC from *T. emersonii*[b] 15 % (w/v), 100 rpm, 50 °C, 72 h Fermentation: *S. cerevisiae*[a], 50 rpm, 38 °C, 72 h **Solid fraction:** SSF: Enzyme filtrase NLC 125 FPU/g and fermentation with *S. cerevisiae*[a], 150 rpm, 38 °C, 72 h	11 % (v/v)	[59]
Olive stone			
1. Acid pretreatment: H_2SO_4 10.5 % (w/v), S/L 33 % (w/v) at 128 °C, 60 min 2. Organosolv pretreatment: Ethanol (50 % w/w), S/L 16.7 % (w/w) at 190 °C, 30 min	**Solid fraction**[b]**:** SSF: Enzyme Cellic Ctec2 (30 g FPU/g) and fermentation with *S. cerevisiae*[a], S/L 20 % w/w, 72 h	47.0 g/L 62.0 g/kg biomass	[41]

(*continued on next page*)

Table 2 (*continued*)

Pretreatment	Hydrolysis and fermentation/Fermentation	Ethanol produced	Ref.
1. Acid pretreatment: H_2SO_4 10.5 % (w/w), S/L 33 % at 180 °C, 60 min 2. Steam explosion pretreatment: 195 °C, 5 min	**Liquid fraction:** Detoxification: 5 % active carbon/liquid, 200 rpm, 30 min, 50 °C Fermentation: *E. coli*[a] SL100, 350 rpm, 37 °C, 96 h **Solid fraction:** Enzymes: Cellic CTec2, 79.9 % (w/w), 50 °C, 72 h Fermentation: *E. coli*[a] SL100, 350 rpm, 37 °C, 70 h	23.5 g/L 160 g/kg biomass	[44]
1. Acid pretreatment: H_2SO_4 10.5 % (w/w biomass), S/L 33 % at 180 °C, 60 min 2. Steam explosion pretreatment: 195 °C, 5 min	**Liquid fraction:** Detoxification conditions: 5 % active carbon/liquid, 200 rpm, 30 min, 50 °C Fermentation: *E. coli*[a] SL100, 350 rpm, 37 °C, 96 h **Solid fraction:** Enzyme: Cellic CTec2, 40 FPU/g of dry substrate, 150 rpm, 50 °C, 240 min Fermentation: *S. cerevisiae*[a], 150 rpm, 35 °C, 48 h	29.9 g/L 162.2 g/kg biomass	[44]
Acid pretreatment: H_2SO_4 1.5 % (w/v), S/L 5 g/100 mL at 121 °C, 60 min	**Solid fraction:** SSF: Enzymes Celluclast and Novozyme 188 and fermentation with *P. tannophilus*[a], S/L 3:1, 100 rpm, 30 °C, 48 h	72.3 g/kg biomass	[60]

FPU: Filter paper unit; S/L: Solid (biomass)/liquid, it refers to solid load or solid-to-liquid ratio, depending on the study; SSF: Simultaneous saccharification and fermentation; IU: International Unit.

[a] C. guilliermondii: Candida guilliermondii; A. niger: Aspergillus niger; T. reesei: Trichoderma reesei; S. cerevisiae; Saccharomyces cerevisiae; S. stipitis: Scheffersomyces stipitis; Escherichia coli: E. coli; T. emersonii: Talaromyces emersonii; P. tannophilus: Pachysolen tannophilus; P. kudriavzevii: Pichia kudriavzevii.

[b] Only the solid fraction recovered after the two-pretreatment step was converted into bioethanol.

Focusing on the former, biogas and biomethane have gained attention as gaseous fuels that are pivotal in transitioning away from fossil fuels. Generating biogas and biomethane from agro-industrial waste is particularly relevant as it contributes to consolidating the circular economy and achieving the Sustainable Development Goals (SDGs) [1]. For biogas to be viable as fuel, the methane (CH_4) content must exceed 50 % [61], while biomethane refers to the upgraded form of biogas, consisting of almost pure methane. An adequate methane content considerably reduces the production of other environmentally harmful gases, such as CO_2 and H_2S, which would otherwise be emitted without proper application. Recently, there has been an increase in studies on the production of biogas and biomethane through anaerobic digestion processes from olive biomass, with promising results being observed (e.g., Table 3). In addition, the use of various types of biomass as a co-substrate for anaerobic digestion has improved production yields [62,63].

Table 3 presents recent studies carried out to obtain biogas using olive biomass as a substrate (mainly olive pomace) or mixtures with other biomasses, such as tomato [64,65] or citrus fruits [65–67] and algae [68–70], to produce synergistic effects. Methane production from olive pomace as a monosubstrate varies between 204 and 418 m^3 CH_4/t volatile solids (VS) [60,70–72], and improvement is achieved by pitting, reaching a production of 660 m^3 CH_4/t VS [73]. The latter study, as exemplified in Fig. 6a, reached a methane purity of about 60 %. On the other hand, the synergistic effects of mixing various agroindustrial biomasses, including olive pomace, which increases the generation of biogas, are due to the most suitable chemical composition for this production. When combined with wheat bran (10 %), methane production

Table 3
Recent studies in methane production to olive biomass: pretreatment, process conditions and yields.

Feedstock Pretreatment	Equipment Inoculum type	Temp HRT ORL	Biogas yield CH_4 yield CH_4 conc.	Ref.
Olive pomace W/o pretreatment	Reactor ($2{\cdot}10^{-3}$ m^3) Mixed culture wastewater treatment plant	35 °C 21 days 1.0 $kgVS^b/m^3{\cdot}d$	– 204 m^3 CH_4/t VS –	[72]
Olive pomace W/o pretreatment	Batch digestor ($1.6{\cdot}10^{-4}$ m^3) Anaerobic sludge biogas plant	37 °C 37 days –	– 255 m^3 CH_4/t VS –	[67]
Olive pomace W/o pretreatment	Reactor ($2.5{\cdot}10^{-4}$ m^3) Industrial up-flow anaerobic sludge blanket reactor	35 °C 28 days –	– 418 m^3 CH_4/t VS –	[70]
Olive pomace W/o pretreatment	Reactor ($5{\cdot}10^{-4}$ m^3)	35 °C 1 day –	– 215 m^3 CH_4/t VS –	[71]
Olive pomace W/o pretreatment	Reactor ($5{\cdot}10^{-4}$ m^3) Cattle manure	37 °C 55 days –	– 233 m^3 CH_4/t VS –	[77]
Olive pomace W/o pretreatment	Reactor ($1.6{\cdot}10^{-4}$ m^3) Mixed culture wastewater treatment plant	35 °C 1 day –	– 150 m^3 CH_4/t VS –	[78]
Pulp and pitted olive pomace W/o pretreatment	Digestor (200 m^3), industrial scale	42 °C 40 days 5.5 $kgVS^b/m^3{\cdot}d$	51 m^3/h 660 m^3 CH_4/t VS 60 %	[73]
Olive stone Olive tree pruning W/o pretreatment	Reactor (18.61 m^3)	35 °C 19 days –	894 m^3/h 325 m^3 CH_4/t biomass –	[46]
50 % Tomato pomace and 50 % olive pomace W/o pretreatment	Bioreactor ($5{\cdot}10^{-3}$ m^3), discontinuous Cow manure	35 °C 25 days –	$4.5{\cdot}10^{-5}$ m^3/h – 65 %	[64]
90 % Cattle manure and 10 % olive pomace W/o pretreatment	Bioreactor ($5{\cdot}10^{-3}$ m^3), continuous	37 °C 15 days –	$1.9{\cdot}10^{-4}$ m^3/h 154 m^3 CH_4/t VS 59 %	[74]
75 % Cattle manure and 25 % olive pomace W/o pretreatment	Bioreactor ($5{\cdot}10^{-3}$ m^3), continuous	37 °C 15 days –	$2.1{\cdot}10^{-4}$ m^3/h 157 m^3 CH_4/t VS 62 %	[74]
60 % Cattle manure and 40 % olive pomace W/o pretreatment	Bioreactor ($5{\cdot}10^{-3}$ m^3), continuous	37 °C 15 days –	$2.1{\cdot}10^{-4}$ m^3/h 151 m^3 CH_4/t VS 63 %	[74]
90 % Pulp pitted olive pomace and 10 % wheat bran shorts W/o pretreatment	Digestor (200 m^3), industrial scale	42 °C 40 days 5.3 $kgVS^b/m^3{\cdot}d$	52 m^3/h 700 m^3 CH_4/t VS 60 %	[73]
20 % Citrus pulp, 13 % poultry manure, 18 % olive pomace, 5 % cattle manure, 20 % whey, 18 % silage, and 6 % tomato peels W/o pretreatment	Semi-continuous reactor ($24{\cdot}10^{-3}$ m^3) Agricultural biogas plant	41 °C 26 days 4.7 $kgVS^b/m^3{\cdot}d$	– 333 m^3 CH_4/t VS 60 %	[66]
75 % Olive pomace, and 25 %	Reactor ($2.5{\cdot}10^{-4}$ m^3) Brewery	– 33 days –	– 441 m^3	[68]

(*continued on next page*)

Table 3 (*continued*)

Feedstock Pretreatment	Equipment Inoculum type	Temp HRT ORL	Biogas yield CH_4 yield CH_4 conc.	Ref.
R. subcapitata[a] W/o pretreatment	industrial up-flow anaerobic sludge blanket reactor		CH_4/t VS 17 %	
75 % Olive pomace, and 25 % *R. okamura*[a] W/o pretreatment	Reactor ($2.5{\cdot}10^{-4}$ m^3) Industrial up-flow anaerobic sludge blanket reactor	35 °C 28 days –	– 454 m^3 CH_4/t VS –	[70]
50 % Olive pomace, and 50 % *C. reinhardtii*[a] *6145* W/o pretreatment	Reactor ($2{\cdot}10^{-4}$ m^3) Industrial up-flow anaerobic sludge blanket reactor	35 °C 29 days –	– 542 m^3 CH_4/t VS –	[69]
62.5 % Corn silage, 12.5 % cattle manure, 12.5 % orange peels, and 12.5 % olive pomace W/o pretreatment	Batch digestor ($1.6{\cdot}10^{-4}$ m^3) Anaerobic sludge biogas plant	37 °C 37 days –	– 317 m^3 CH_4/t VS –	[67]
Olive pomace NaOH 8 % w/w, 25 °C, 1 d	Reactor ($5{\cdot}10^{-4}$ m^3)	35 °C 1 day –	– 280 m^3 CH_4/t VS –	[71]
Olive stone NaOH 1.5 % w/w, 121 °C, 1 h	Reactor ($1.6{\cdot}10^{-4}$ m^3) Mixed culture wastewater treatment plant	35 °C 15 days –	– 252 m^3 CH_4/t VS –	[60]

Conc. = concentration; HTR=Hydraulic retention time; OLR: Organic loading rates; Temp = temperature; VS: Volatile solid.

[a] C. reinhardtii: Chlamydomonas reinhardtii; R. subcapitata: Raphidocelis subcapitata; R. okamurae: Rugulopteryx okamurae.

increases, reaching up to 700 m^3 CH_4/t VS [73]. According to Ref. [74], adding olive pomace at a rate higher than 40 % reduces biogas and methane production.

Mixtures involving more than one type of biomass benefit from synergistic effects [67], increasing methane production by 20 % and achieving 317 m^3 CH_4/t VS [65,66].

The combination with aquatic biomasses, such as *Chlamydomonas reinhardtii* 6145, is also interesting as methane production of 542 m^3 CH_4/t VS has been achieved in 50 % of mixtures [69]. Other studies were done with *Raphidocelis subcapitata* and *Rugulopteryx okamurae*, with slightly lower methane production, 441 and 454 m^3 CH_4/t VS, respectively [68,70].

Overall, it is important to note that the observed increase in methane yield from biomass mixtures may not solely result from the contribution of olive biomass; instead, it could be significantly influenced by the biogas potential of the other co-substrates used in the mixture.

Other studies have applied a pretreatment on olive pomace and olive stone before digestion. The pretreatment of olive pomace does not significantly improve biomethane production [60,71,75]. Furthermore, other studies (olive pomace, tomato peel, etc.) have revealed that an extraction step before pyrolysis can favor the energy of the gas obtained compared to untreated biomass, although the effectiveness depends on the specific feedstock [76]. A photochemical pretreatment with olive pomace was also carried out, but no improvement in biogas or methane production was observed [77].

5.2.2. Hydrogen

Following gaseous fuels as targeted products, recent works highlight dark fermentation and thermochemical conversion by gasification and pyrolysis. Hydrogen is produced from biomass through these conversion routes. When hydrogen is generated using biological methods, it is referred to as biohydrogen. In dark fermentation, hydrogen yields ranging between 0.12 and 0.29 m^3 H_2/kg of glucose have been reported, with the highest yield obtained from nanoparticles synthesized with olive leaves as a reducing agent. Notably, using $NiFe_2O_4$ nanoparticles outperformed $CoFe_2O_4$ in hydrogen production. It has been suggested that nanoparticles can increase bacterial hydrogenase and favor the catalytic reduction of protons to molecular hydrogen [79]. In the dark fermentation process, pretreatments with oxalic or sulfuric acid achieved hydrogen yields of 0.12 and 0.13 m^3 H_2/kg of glucose, respectively [80] (Table 4). This study has been exemplified in Fig. 6b, which does not significantly differ from or improve upon the yield obtained using nanoparticles. This underscores the importance of further research into nanoparticle applications in dark fermentation, as they achieve better yields and utilize more environmentally friendly metal nanoparticles. Integrating nanoparticles with olive biomass appears to be a promising strategy to enhance hydrogen production efficiency.

At the level of gasification processes, a hydrogen production of 32.7 m^3 H_2/h was achieved using a rising bubbling bed reactor gasifying olive pomace [81]. This study has been exemplified in Fig. 6c. This production, achieved on a pilot scale, represents a considerably higher rate compared with other published studies using the same type of biomass [81–83]. However, mixing olive pomace with wastewater as a gasification substrate does not considerably improve hydrogen production, as reported in Ref. [82], where a maximum yield of $8.2{\cdot}10^{-4}$ m^3 of H_2 was achieved at 900 °C (Table 5). Integrating olive pomace in gasification processes remains a promising strategy to enhance hydrogen production efficiency.

In the case of olive pruning gasification, using a radio frequency plasma torch (RFPT) treatment increases hydrogen production from 0.17 m^3 H_2/h at the gasifier outlet to 0.65 m^3 H_2/h after passing through the torch. This is attributed to the elevated plasma temperature, which promotes steam and dry reforming reactions, resulting in higher hydrogen production and an increased hydrogen-to-carbon dioxide ratio, from 0.73 to 1.5, as well as a reduction in benzene/toluene levels [84]. It will be essential to investigate the application of this technology to other types of olive-derived biomasses to determine whether similar production levels can be maintained.

The pyrolysis process for olive pomace has shown that hydrogen production significantly benefits from using catalysts (Table 5). As demonstrated by Ref. [85], hydrogen yields increased by 70 % when a Ni-Co/Al2O3 catalyst was applied, indicating a substantial improvement compared to gasification under similar conditions. In addition to hydrogen, pyrolysis produces biochar as a valuable by-product, which has potential applications in other fields, such as the removal of heavy metals like lead [86], nitrogen fertilizer fixation [87] or as a carbon removal strategy [88]. These complementary uses of biochar further enhance the sustainability and economic feasibility of pyrolysis processes for olive biomass.

6. Bioactive compounds and platform chemicals from olive biomass

In the context of biomass and biorefineries, a building block refers to key chemical compounds that serve as a fundamental precursor for synthesizing a wide range of value-added products [89]. Numerous pathways for obtaining these products have been proposed, depending on the starting raw material (summarized in Fig. 4). In the case of olive biomass, the range of products that can be obtained is extensive, generally through the fermentation of the sugars contained in the cellulosic and hemicellulosic fractions. Another group of valuable compounds can be obtained through extraction, typically from the non-structural fractions of the biomass. Lastly, the lignin fraction also serves as a source of a series of compounds, further expanding the portfolio of products derived from olive biomass, Fig. 7.

For example, a recent study showed the potential of olive leaves to be converted into various products, assessing the feasibility of producing

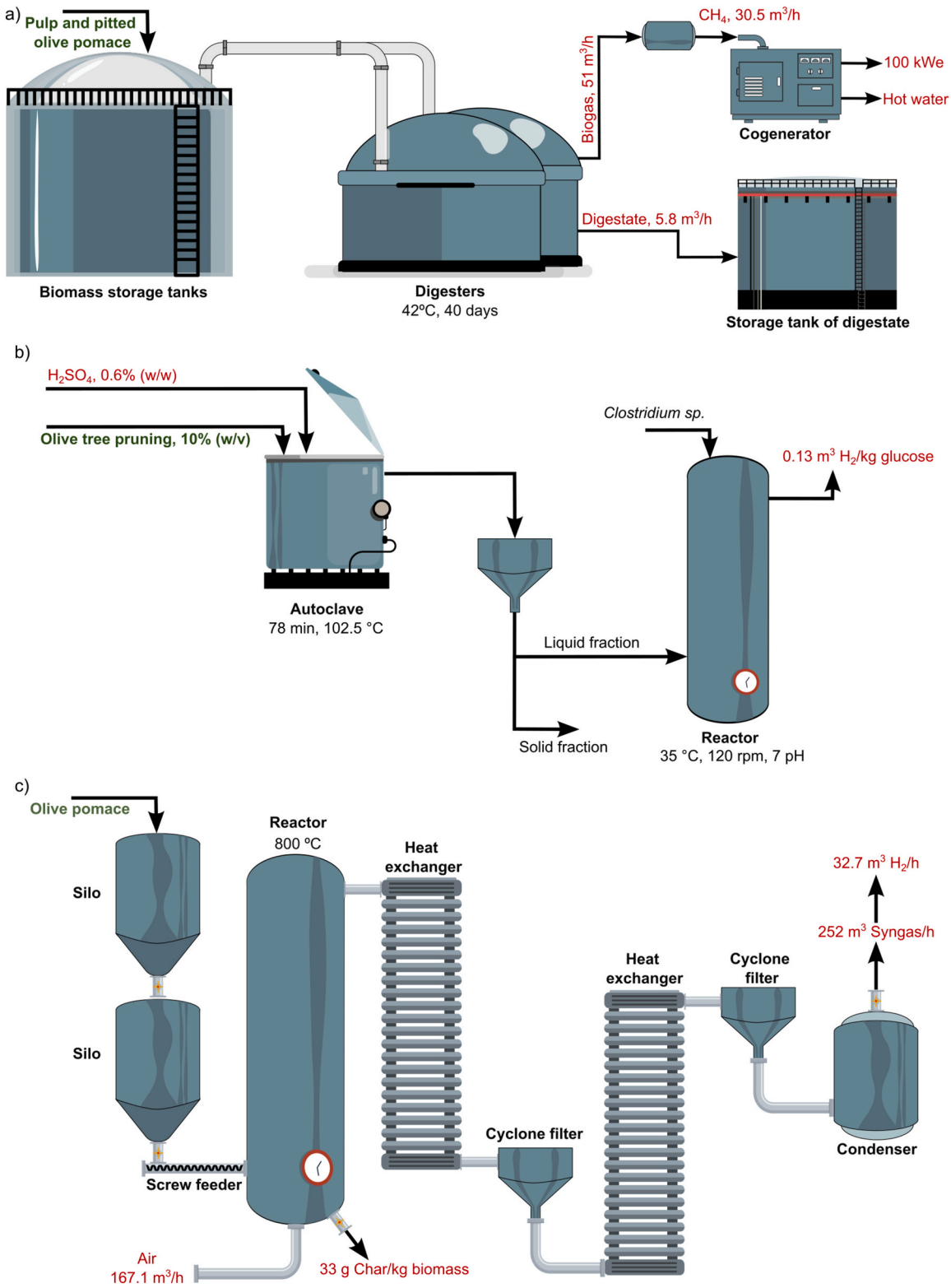


Fig. 6. Representation of studies on the valorization of the biomass olive oil industries: a) Biogas and methane from pulp and pitted olive pomace [73]; b) hydrogen production from olive tree pruning [80]; c) syngas and hydrogen from olive pomace [81].

antioxidants, bioethanol, and organosolv lignin, along with the techno-economic and environmental performance of such a biorefinery [90]. The study concluded that producing 166 kg of antioxidant extract, 111 kg of organosolv lignin, and 28 kg of ethanol per ton of olive leaves is feasible at a processing scale of 30,000 t/yr. The carbon footprint of the biorefinery was 1.13 kg CO_2-eq/kg olive leaves. Another study evaluated two biorefinery schemes using 100 t/day olive tree pruning waste. The first scenario proposed the production of ethanol (6363.6 m^3/yr), xylitol (999.3 t/yr), and antioxidants (1221.5 t/yr), while the second scenario focused on the standalone production of antioxidants. Both scenarios showed similar economic viability (net present value of 11.9 M€); however, the second scenario reduced key environmental

Table 4
Recent studies in biohydrogen production to olive biomass by dark fermentation: pretreatment, process conditions, and yields.

Feedstock Pretretment	Process conditions	Production of H_2	Ref.
Olive leaves W/o pretreatment	*Clostridium* sp. (10 % v/v) 200 mg/L $NiFe_2O_4$ 37 °C, 130 rpm	0.29 m^3 H_2/kg glucose[a]	[79]
Olive leaves W/o pretreatment	*Clostridium* sp. (10 % v/v) 200 mg/L $CoFe_2O_4$ 37 °C, 130 rpm	0.27 m^3 H_2/kg glucose[a]	[79]
Olive tree pruning Oxalic acid at 10.5 % (w/w), 36 min, 114 °C	*Clostridium* sp. 35 °C, 120 rpm, 7 pH	0.12 m^3 H_2/kg glucose[a]	[80]
Olive tree pruning Sulfuric acid at0.6 % (w/w), 78 min, 102.5 °C	*Clostridium* sp. 35 °C, 120 rpm, 7 pH	0.13 m^3 H_2/kg glucose[a]	[80]

[a] Recalculated considering T = 35 °C, P = 1 atm and glucose = reducing sugar.

impacts, such as the global warming potential, compared to the multi-product scheme [91]. Another study by Ref. [89] focused on olive tree prunings, alongside residues from pine and eucalyptus, prominent examples of residual biomass in Spain. Several building blocks were evaluated, including xylitol, sorbitol, succinic, glutamic, glucaric, levulinic, lactic, and itaconic acids, from a short-term economic potential perspective. The study highlighted that incorporating building blocks is crucial to improving the profitability of biorefineries.

Moving to industrial olive biomasses, olive stones have been explored as valuable raw materials [41]. proposed the production of furfural, lignin, antioxidants, and sugars using a combination of pretreatments to enhance the sugar yields. This approach achieved up to 70 % sugar release from the hemicellulose fraction, combined with 41 % glucose yield from enzymatic hydrolysis of the pretreated solids, along with 76.3 % lignin recovery and 46 % furfural yield [90]. In a recent study, the comprehensive valorization of olive stones, focusing on their potential as a biofuel while simultaneously investigating the extraction of valuable antioxidants, was reported [92]. The study evaluated the efficacy of different extraction solvents on antioxidant activity and assessed the stability of these antioxidants after in vitro gastrointestinal digestion.

The use of olive pomace as a feedstock for a biorefinery has been recently reviewed by Ref. [4]. In addition to presenting the utilization pathways for obtaining the main products, including sugars and their derivatives (including polyols, oligosaccharides, and pectins), phenolic compounds, and triterpenes, among others, this review critically analyzes the contribution that these biorefineries can have to the achievement of the SDGs and the development of the circular bioeconomy. Recently, the valorization of oven-dried olive pomace, from both two-phase and three-phase olive oil mills, for potential applications in food, nutraceutical, and cosmetic products was reported. The research evaluated the antioxidant and enzyme inhibitory properties of the olive pomace, highlighting its potential for health-promoting and cosmetic uses [93].

A selection of recent papers dealing with bio-based chemicals is summarized in Table 6. Among them, polyphenols constitute a broad group of compounds that are receiving increasing attention lately due to their potential applications, especially health-related ones, and as antioxidants [94].

7. Bio-based materials from olive biomass

The traditional use of olive stones, exhausted olive pomace, and the woody fraction of olive tree pruning has been used as solid fuels in heating systems or to generate electricity [4]. Additionally, olive pomace has been studied mainly for compost production and the generation of solid and liquid organic fertilizers [105]. However, these

Table 5
Recent studies in hydrogen production to olive biomass by gasification and pyrolisis: process conditions and yields.

Feedstock Equipment	Process conditions	Production of H_2 Other coproducts	Ref.
Gasification			
Olive tree pruning Rotary kiln gasifier with radio frequency plasma torch	Air, 1 rotations/min 0.24–0.6 m^3 air/h 450 g/h of biomass 650 °C	0.17 m^3 H_2/h[a] 0.65 m^3 H_2/h after torch[a]	[84]
Olive pomace Pilot-scale fixed-bed downdraft reactor (1.5 m height, 0.4 m width)	Air, 29.84 m^3/h 800 °C, 420 min	7.8 m^3 H_2/h 47 m^3 Syngas/h 55 g Char/kg biomass	[81]
Olive pomace Bubbling fluidized-bed (0.5 m internal diameter, 4.15 m high)	Air, 167.12 m^3/h 800 °C, 420 min	32.7 m^3 H_2/h 252 m^3 Syngas/h 33 g Char/kg biomass	[81]
Olive pomace Laboratory scale fixed bed reactor	Oxygen/stem ratio 0.5 $2.16{\cdot}10^{-5}$ m^3/h 10 g biomass 700 °C	$8.9{\cdot}10^{-4}$ m^3 H_2[b]	[82]
Olive pomace Laboratory scale fixed bed reactor	Oxygen/stem ratio 0.5 $2.16{\cdot}10^{-5}$ m^3/h 10 g biomass 900 °C	$9.1{\cdot}10^{-4}$ m^3 H_2[b]	[82]
80 % Olive pomace and 20 % olive waste water Laboratory scale fixed bed reactor	Oxygen/stem ratio 0.5 $2.16{\cdot}10^{-5}$ m^3/h 10 g biomass 700 °C	$7.8{\cdot}10^{-4}$ m^3 H_2[b]	[82]
80 % Olive pomace and 20 % olive waste water Laboratory scale fixed bed reactor	Oxygen/stem ratio 0.5 $2.16{\cdot}10^{-5}$ m^3/h 10 g biomass 900 °C	$8.2{\cdot}10^{-4}$ m^3 H_2[b]	[82]
Olive pomace Fixed bed reactor 3 L (updraft reactor)	Dry air, $5{\cdot}10^{-5}$ m^3/h 75 g of biomass 900 °C	$2.4{\cdot}10^{-5}$ m^3 H_2/h	[83]
Olive pomace Fixed bed reactor 3 L (downdraft reactor)	Dry air, $5{\cdot}10^{-5}$ m^3/h 75 g of biomass 800 °C	$2.3{\cdot}10^{-5}$ m^3 H_2/h	[83]
Pyrolysis			
Olive pomace Fixed-bed updraft reactor, heated by electrical resistance	N_2: 0.18 m^3/h, 20 °C, 1 atm 3 g of biomass 700 °C, 15 min	0.14 m^3 H_2/kg biomass[a]	[85]
Olive pomace Fixed-bed updraft reactor, heated by electrical resistance	N_2: 0.18 m^3/h, 20 °C, 1 atm 3 g of biomass Catalyst: Ni–Co/Al_2O_3, 20 % (w/w) 700 °C, 15 min	69.9 (%w/w) syngas (0.49 m^3 H_2/kg biomass[c]) 22.2 (%w/w) char 5.4 (%w/w) bio-oil	[85]

[a] Recalculated considering P = 1 atm; T = 650 °C.
[b] Total volume; the unit time was not specified.
[c] Recalculated considering P = 1 atm; T = 700 °C. The unit time was not specified.

applications fail to fully explore the potential value of olive biomass and provide a limited economic return for farmers, e.g., 0.03 €/kg of wood from olive tree pruning, while exhausted olive pomace costs around 0.012–0.045 €/kg [106,107].

Beyond these traditional uses, recent research highlights the versatility of olive biomass in producing a diverse range of high-value materials. For instance, residues from olive cultivation and olive oil production can be effectively recycled into ceramic materials, offering advantages such as reduced greenhouse gas emissions and improved fuel

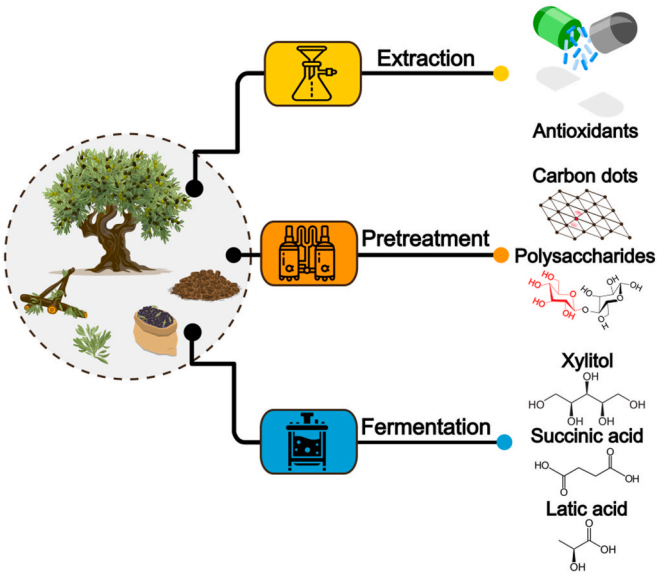


Fig. 7. Main biobased chemicals derived from olive biomass.

efficiency during manufacturing processes [108]. Furthermore, olive stone waste has been successfully incorporated as a natural filler in polymeric composites, specifically with recycled high-density polyethylene (rHDPE), demonstrating improved material properties and promoting a circular economy model [109].

Alternatively, these olive biomasses can serve as feedstock for producing various high-value materials, thereby increasing its economic and industrial potential. Key examples include activated carbon [110], biochar [111], biopolymers [112], carbon dots [42], cellulose nanofiber [42,113], nanocrystals [114], and reinforcement materials [115].

Biochar has been receiving significant attention in recent years as a versatile material with numerous advantages and applications. Valorizing olive biomass into biochar, the carbon-rich product derived from pyrolysis, is particularly promising. Biochar can be used as an adsorbent of toxic compounds and as a soil amendment in agriculture [110]. The physicochemical properties of the biochar can be further enhanced through activation, opening new possibilities for its use. The production of activated carbons has been proposed as an efficient transformation of olive pomace. Unlike conventional biochar, activated carbons have a much higher specific surface area, making them ideal for liquid and gas adsorption treatments, and carbon-capture materials. For example [116], demonstrated that the surface area of olive pomace biochar can be increased by washing with water from 26 to 732 m^2/g, and up to 1600 m^2/g when activated with KOH and washed with an acid solution. Similarly [110], prepared activated carbons with high surface area by KOH activation, estimating a production cost of about 9.3 €/kg, which positions these biocarbons as competitive with commercial activated carbons priced at 9.2–10.5 €/kg in Spain [117].

In a different application, olive mill wastewater, traditionally a disposal challenge, can be transformed into bio-lubricants using innovative enzyme-immobilized nanocatalysts, thus repurposing valuable residual oils and offering sustainable alternatives to fossil-based lubricants [118]. Additionally, olive pomace-derived hard carbons are being explored as high-performance, cost-efficient negative electrode materials for sodium-ion batteries, with studies showing excellent capacity retention and rate capabilities after specific chemical activation and carbonization processes [119].

Lignin from biorefinery streams also has the potential to be valorized, tailoring its properties by the treatment applied for its valorization. Pyrolysis and hydrothermal carbonization, combined with physical and chemical activation, can be applied to obtain activated carbon from lignin [7]. These approaches demonstrate the feasibility of transforming

Table 6
Bio-based chemicals derived from olive biomass in recent studies.

Bio-based compound	Primary source and comments	Ref.
Bioactive compounds		
Bioactive compounds	Bark from olive tree pruning. Hexylresorcinol, 1-methyl-N-vanillyl-2-phenethamine, and allopurinol were identified in 25-year-old olive tree pruning residues	[95]
Phenolic rich extracts	Olive leaves and olive mill wastewater. Phenolic-rich extracts (500 and 1000 mg/kg) from olive leaves and olive mill wastewater were added to conventional gluten-free formulations as a potential strategy for creating nutritious and healthy breadsticks with extended shelf life	[96]
Hydroxytyrosol	Supercritical CO_2 extraction of olive pomace. The anti-inflammatory and immunomodulatory activities against dry eye disease were evidenced	[94]
Polyphenols, lipids and algal biomass	Olive pomace and table olive processing wastewater. 54.5 kg of sugars and 3.2 kg of polyphenols were released from 100 kg of exhausted olive pomace. After the recovery of polyphenols, microalgae used hydrolysates to produce lipids	[97]
Hydroxytyrosol and fiber	Olive pomace. Yogurt was supplemented with pulp-enriched powder obtained from olive pomace, improving healthy properties	[98]
Antioxidants	Crushed stones, olive pomace, defatted olive pomace, olive leaves Supercritical CO_2 extraction was used to determine antioxidant compounds, including hydroxytyrosol, chlorogenic acid, caffeic acid and ferulic acid; best conditions were compared to conventional extraction	[36]
Antioxidants	Dried samples (50–100 °C) showed high fiber (45–66 %), protein (5–10 %), and lipid (17–21 %, mainly oleic acid) contents, with antioxidant and enzyme inhibition activities suggesting potential nutraceutical and cosmetic applications.	[93]
Other compounds		
Fatty acid methyl esters	Alperujo. The production of biodiesel from fatty acid methyl esters of residual olive pomace oil obtained by supercritical fluid extraction and Novozym 435 and Lipozyme TL IM 10 % as catalyst was studied. The process achieved 96 % conversion to methyl esters.	[40]
Polygalacturonases and omega-3 fatty acids	Extracted olive pomace. A cost-effective process for producing pectinases and omega-e3-enriched lipids is proposed using *C. cohnii*[a]	[99]
Oligosaccharides	Extracted olive pomace. Hydrothermal treatment was applied to extract olive pomace coming out from different olive varieties. Maximum yields of oligosaccharides of 98 % were obtained at a severity factor of 4.0	[100]
Furfural	Olive stones. Furfural at a 63.3 % yield and 18 g/L concentration was produced from olive stones subjected to an autoclave pretreatment followed by 2.6 %w/v H_2SO_4 treatment in a microwave reactor at 200 °C, equivalent to almost 90 kg of furfural per tonne of processed stones	[41]
Succinic acid	Olive stones. Xylose-rich hemicellulosic hydrolysates of olive stones were used to produce succinic acid by *A. succinogenes*[a], which was separated by direct crystallization method	[45]
Lactic acid	Olive stones. Dilute acid hydrolysis was applied to generate a xylose-rich solution that was further fermented to lactic acid by *B. coagulans*[a] under non-sterile conditions	[101]

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Table 6 (*continued*)

Bio-based compound	Primary source and comments	Ref.
Xylitol	Olive stones. A two-step pretreatment using water and dilute acid solutions was applied to olive stones to produce a 60 g xylose/L solution, which was detoxified and further converted into xylitol by *C. boidinii*[a]. Pure xylitol crystals with a yield equivalent to 12 g xylitol per 100 g olive stones were obtained	[102]
Fermentable sugars	Olive stones. A two-step sequential fractionation process of olive stones was studied, consisting of a first dilute sulfuric acid treatment at 128 °C, yielding a 71 % xylose recovery; the resulting solid was submitted to steam explosion at optimized conditions. Globally, 83 % of the sugar content of olive stones was available for further conversion	[32]
Lignin with antioxidant properties	Olive tree pruning. Deep eutectic solvents were used to extract lignin with high purity (>75 %)	[104]
Rhamnolipids	Olive mill wastewater. When olive mill wastewater was supplemented with corn-steep liquor, an increase in production of rhamnolipids by *B. thailandensis*[a] was detected	[103]

[a] *C. cohnii: Crypthecodinium cohnii; A. succinogenes: Actinobacillus succinogenes; B. coagulans: Bacillus coagulans; C. boidinii: Candida boidinii; B. thailandensis: Burkholderia thailandensis.*

low-value olive biomass fractions into competitive, high-value products.

Another promising valorization pathway involves the production of biodegradable biopolymers, creating new opportunities for more sustainable applications in the food, agriculture, and medical fields [120]. For example, bioplastics derived from hemicellulose in olive pomace have been developed using renewable polyols such as glycerol, sorbitol, and xylitol as plasticizers. This approach enabled the formulation of flexible and biodegradable films with barrier properties and varying tensile strengths [112]. Rodríguez-Liébana et al. have developed biocomposites based on polylactic acid derived from olive tree pruning cellulose [121]. Another possible route is fermentation, for example, using *Bacillus amyloliquefaciens* and olive mill wastewater as a carbon source (50 %) to produce polyhydroxyalkanoate (PHA) bioplastic [122]. PHA-based plastics are biocompatible and biodegradable, making them suitable for various sectors. Moreover, using low-cost liquid waste effluents as carbon sources opens the door to minimizing PHA production costs, which remains one of the primary challenges for the commercial viability of this PHA conversion route [123].

The obtainment and functionalization of carbon dots represent a rapidly expanding field of research, offering exciting possibilities across various applications. These nanomaterials, composed of carbon atoms with sizes typically below 10 nm, have demonstrated potential as antimicrobials [124], and also have applications in fields such as bioimaging, sensors, photocatalysis, drug delivery, and light-emitting devices [125]. Carbon dots can be derived from olive biomass, such as olive stones, through hydrothermal treatment followed by acid treatment [42]. Additionally, hydrothermal carbonization has been applied to olive pomace as feedstock for producing carbon dots [126].

Another material of interest is cellulose nanofibers, which refer to flexible elementary nanofibrils of 5–50 nm obtained from cellulose fibers [114]. The composition and properties of cellulose-based nanoparticles are highly influenced by the type of pretreatment (Table 7). For example, a cellulose nanocrystal fraction obtained from olive leaves had a higher reinforcing ability than those containing cellulose nanofibrils and lignin-cellulose nanofibrils. This finding was attributed to the higher crystallinity degree of this fraction and the needle-like form of the cellulose nanoparticles [114]. Additionally, their visco-elastic gel properties make them suitable for a wide range of applications [42].

Table 7
Recent studies on bio-based materials produced from olive-derived biomass.

Bio-based material	Comments	Ref.
Olive tree pruning		
Cellulose nanofiber	Kraft pulping followed by oxygen delignification was applied on olive tree pruning for delignification. Then, the pretreated solid was treated by (2,2,6,6-tetramethylpiperidine-1-oxyl radical) (TEMPO)-mediated oxidation and defibrillated to obtain cellulose nanofiber. It showed good visco-elastic gel properties	[42]
Reinforcement material	Olive tree pruning was treated with 10 % w/w Na_2SiO_3, 3 % w/w $CaCl_2$, 5 % w/w NaOH, and 6 % w/w silane in water. The fiber obtained by sodium silicate improved flexural and compressive strength, strength values compared to unreinforced control, and decreased conductivity	[115]
Biochar-fertilizers	The biochar of olive tree pruning is obtained at 400 °C, and HNO_3 is activated. This is used as biochar-based fertilizers (BBFs) with increased N concentration	[87]
Olive leaves		
Cellulose and cellulose-rich nanoparticles	Cellulose nanoparticles were obtained using different pretreatment configurations on extracted olive leaves: 1) cellulose nanocrystals: alkaline, bleaching, and acid treatments; 2) cellulose nanofibrils: bleaching, TEMPO oxidation, and high-pressure homogenization; 3) lignin-cellulose nanofibrils: TEMPO oxidation and homogenization; 4) lignin-cellulose nanofibrils: bleaching and high-pressure homogenization	[114]
Olive pomace		
Activated carbon	Activated carbons were obtained through mechanical pre-activation with KOH, pyrolysis t(~900 °C), and chemical purification. The activated carbons had high specific areas (>1600 m^2/g) and carbon content (>90 %) was obtained. The volatile organic components were reduced using the latter treatment	[110]
Activated carbon	Olive pomace biochar was activated with KOH at 750 °C in an inert atmosphere to be used as CO_2 adsorbent. The surface area was 1601 m^2/g	[116]
Hemicelluse-based bioplastic	Hemicellulose was solubilized by KOH at room temperature and precipitated by acetic acid:ethanol 1:10. The hemicellulose was mixed with various polyols plasticizers at 90 °C to obtain films by solvent casting method. Glycerol-hemicellulose bioplastics presented low solubility and high-water vapor permeability for food application packaging, while xylitol-hemicellulose bioplastics had higher mechanical strength	[112]
Biochar	Olive pomace biochar obtained at 500 °C was used to remove lead from aqueous solutions	[86]
Biochar	Physical and chemical activation of olive pomace biochar for CO_2 capture. KOH activation significantly enhances microporosity and CO_2 adsorption capacity, making olive pomace a promising precursor	[127]
Hydrocarbons	The production of enhanced hydrocarbons through the recirculation of the aqueous phase from the hydrothermal carbonization (HTC) process of TPOMW as a reactive medium contributes to the recovery of valuable organic compounds that can be utilized as precursors for new materials in various industrial applications	[128]
Olive stone		

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Table 7 (*continued*)

Bio-based material	Comments	Ref.
Carbon dots	Hydrothermal pretreatment was applied to olive stones to produce a xylose-rich hydrolysate, which was further submitted for a one-step treatment in mild conditions with trifluoroacetic acid and different solid catalysts. As a result, interesting fluorescent nanosized (4.5 nm) particles were obtained, displaying the characteristic behavior of carbon dots	[42]
Olive mill wastewater		
Polyhydroxyalkanoate bioplastic	Polyhydroxyalkanoate (1.45 g/L) was produced using *B. amyloliquefaciens*[a] OM81 in the presence of 50 % olive mill wastewater and recovered by chloroform	[122]
Bio-lubricants	An innovative method for bio-lubricant production from olive mill wastewater using enzyme-immobilized nanocatalysts. Repurposes valuable residual oils from olive mill wastewater, offering a sustainable alternative to fossil-based lubricants	[118]

[a] *B. amyloliquefaciens*: *Bacillus amyloliquefaciens*.

Olive biomass fibers and derived fly ash can also be applied to formulate new material products or to modify the properties of cement and clay ceramic bricks, contributing to sustainability in the building sectors and cost reduction [6,115,129,130]. For example, olive tree pruning fibers have been used to improve the properties of alkali-activated materials, such as flexural and compression strength, depending on the treatment and cellulose content. At the same time, thermal properties like conductivity were decreased. These findings are noteworthy because alkali-activated cement reinforced with olive tree pruning fibers offers a sustainable alternative to traditional Portland cement [115]. Other studies showed that olive tree pruning treated with Na_2SiO_3 can be utilized to produce indoor covering material with low thermal conductivity and adequate hygrothermal and acoustic performance [129].

All these olive-based materials can be produced as stand-alone processes or through processes integrated into a biorefinery, enabling the development of new types of industries in the olive sector. For example, previous research applied a multiproduct scheme to olive leaves, involving an extraction step to recover bioactive compounds, followed by producing nanocellulose particle types from the extracted solid [114]. This approach allowed for fractionating the extracted solid, resulting in the recovery of cellulose nanocrystals, cellulose nanofibrils, and lignin-containing cellulose nanofibrils with different reinforcing properties in nanocomposite films. In another study, an instant controlled pressure drop was applied to obtain phenolic compounds from olive pomace, and the resulting solid was treated with KOH to solubilize the hemicellulose, which was recovered to formulate bioplastics [112]. In a techno-economic study, another work evaluated the integration of biogas and PHA production using olive mill wastewater as feedstock. This study demonstrated the potential for generating profits of approximately 350,000 € over five years, depending on the amount of subsidies [131].

8. Current challenges and future perspectives in olive biomass biorefineries

Despite the abundance and versatility of olive-derived biomass, industrial-scale biorefineries remain limited due to technological, economic, and regulatory challenges. Only a few energy-driven biorefineries utilizing olive-derived biomass are operational [132,133], generally restricted to small-scale plants near olive oil mills and extractive facilities, converting olive pomace into heat, electricity, or district heating via combustion or gasification [132]. Recently, the valorization of olive tree prunings into biochar has gained interest for its carbon removal potential and diverse applications [134].

The small scale of current facilities is largely due to logistical and technical barriers, such as high moisture content, chemical heterogeneity, and seasonal availability of olive biomass, which hinder transportation and value chains [135]. These challenges impact both economic and environmental performance. Ongoing research emphasizes the development of adaptable pretreatment and processing technologies to mitigate these limitations and enable large-scale implementation. Effective pretreatment remains essential for improving biomass conversion, yet conventional methods (mechanical, chemical, and thermal) are often energy-intensive and costly, thereby limiting the investment [32]. Emerging eco-friendly alternatives like deep eutectic solvents (DES) and NADES show promise [20], enabling lignocellulosic breakdown under milder conditions and thus improving operational costs and sustainability [136]. NADES have successfully extracted bioactive compounds such as hydroxytyrosol from olive leaves and pomace [136–138] and have been explored for polyphenol recovery [139,140]. However, despite their potential, further research is needed to address scalability and economic viability. For example, operational issues that need to be addressed include mass transfer problems caused by viscosity, purity, and DES recycling, among others. However, ready-to-use formulations in the pharmaceutical and cosmetic fields could be developed for bioactive compound valorization, avoiding some of these issues [20,139].

Another option to face high initial investments and operational costs [141] is diversifying outputs through cascading schemes, coupling the production of energy products with compounds with high market value—e.g., xylitol, succinic acid, antioxidants, etc.—can improve profitability [44]. Studies reveal that profitability can be achieved within 5–7 years, especially when antioxidants are included [91,142] or studies that evaluate thermochemical processes, gasification, or pyrolysis of olive pomace reach 4–5 years [143]. Steam explosion pretreatment outperforms organosolv methods economically [91]. Integrating cascading processes enhances resource efficiency while offering sustainable solutions for agricultural waste management in regions like Southern Europe [91,144]. For example, a plant processing 8000 t/year of olive pomace may reach payback in ~8 years, though profitability depends on antioxidant and solvent prices [145].

Despite progress, biorefineries still face challenges in competing economically with the petrochemical industry, which benefits from scale and existing infrastructure. Low consumer awareness and limited demand for bio-based products further hinder market development. Improving economic performance requires multi-product strategies, such as upgrading bioethanol to higher-value chemicals, and adopting advanced modeling tools to optimize processes and minimize waste. Moreover, robust catalysts and genetically engineered microorganisms are essential to improving conversion efficiency and reducing costs [8].

The environmental sustainability of olive biomass biorefineries must be prioritized [8]. Although aligned with circular economy principles, these processes must demonstrate tangible environmental benefits where life cycle assessment (LCA) methodology plays a central role. For instance, anaerobic digestion of olive oil waste showed lower impacts compared to traditional disposal [146,147]. LCA has also been assessed for the valorization of olive pomace—particularly its stone fraction—as reinforcement in polypropylene and polyethylene biocomposite materials [148], as well as in renewable energy applications [149]. Gasification of pruning residues for biochar and tar applications add further environmental savings [90,148,149]. Among thermochemical options, LCA ranks combustion as most impactful (100 %), followed by gasification (69.45 %), HTC (50.96 %), and pyrolysis (40.97 %). Heat recovery during drying can reduce these impacts by up to 36.70 % [143].

Combining LCA with techno-economic assessments provides valuable insights into environmental, technical, and economic performance. Most studies using this strategy have adopted a cradle-to-gate approach to conduct the LCA, where the system boundaries extend from raw

material extraction to the product leaving the biorefinery based on olive biomass (Fig. 8) [90,91,149,150]. In these studies, the allocation method—how to distribute environmental burdens—is not uniform, as it is based on mass allocation (e.g., the quantity of bioproducts produced) [89,145], whereas in others, it is based on economic considerations [149,150].

Cradle-to-gate approach was followed to assess the co-production of antioxidants, lignin, and bioethanol from olive leaves [90], and antioxidant production from olive tree pruning [91], demonstrating their potential to reduce impacts, manage waste, and generate value. Nevertheless, achieving true sustainability requires holistic evaluation, encompassing economic, environmental, and social criteria, with a focus on resource utilization, enhancing operational efficiency, and broadening market opportunities in olive-based systems [151,152]. Additionally, adopting a cradle-to-grave approach—which considers all stages from raw material extraction to use and end-of-life disposal—supports decision-making by identifying cases where using an environmentally friendly process might inadvertently increase environmental impacts due to the use of bio-based products (Fig. 8) [153].

An emerging area of research in olive-derived biomass biorefineries involves integrating Carbon Capture and Storage (CCS) systems [88]. CCS allows for capturing biogenic carbon dioxide (CO_2) from processes such as combustion, gasification, or fermentation with subsequent storage in geological formations or utilization in industrial applications (Carbon Capture and Utilization, CCU). When properly managed, such systems can result in carbon-negative biorefineries, producing olive-based products while removing more CO_2 from the atmosphere than they emit over their life cycle [88]. Studies have shown that incorporating BECCS using olive tree pruning to power the olive mill operation can produce carbon-negative virgin olive oil. Patzsche et al. simulated a 100-MW solar-powered thermo-chemical air separation system combined with an oxy-fuel power plant, capturing 260.5 kt of CO_2 annually using pruning residues [154]. However, collateral damages must be considered [88]. For example, a recent study on a the biorefinery installation utilizing olive pruning as raw material has found that the use of BECCS reduces the carbon footprint, offering potential through premium markets for carbon-negative products and removal credits (i.e., −84.4 kg CO_2eq per 1 kg of bioethanol and 0.15 kg of antioxidants). However, these environmental benefits are offset by trade-offs such as increased eutrophication (+75 %), water consumption (+45 %), soil impacts (e.g., +80 % of land use), and acidification (+107 %) [150,150,154]. However, traditional practices, such as like burning or soil incorporation of prunings, also carry environmental impacts.

Valorizing these residues transforms waste into valuable bioenergy and products, reducing reliance on fossil fuels and CO_2 emissions, while providing social benefits. For farmers and the olive oil industries, it creates an additional income stream, fosters job creation across the supply chain, and promotes rural development and economic resilience. This is in line with a recent social LCA study conducted in the Netherlands that suggests biorefinery systems can create reliable employment opportunities and attract high-quality human resources [155]. Moreover, by converting agricultural by-products into sustainable bioproducts, they foster clean technology innovation [156]. However, their large-scale deployment still faces challenges related to market acceptance, feedstock availability, and techno-economic feasibility [157]. Regarding CCS scalability, mature technologies such as amine-based absorption are scalable in the short term, but remain energy-intensive. Emerging technologies, such as calcium looping, hold significant potential but require further development before they can be commercialized on a large scale [158].

Overcoming the challenges of olive biomass valorization requires supportive policies, targeted R&D investments, and a strong regulatory framework. Incentives such as subsidies or tax breaks can improve the economic viability of biorefinery projects. Increased R&D funding is key to enhancing process efficiency, lowering costs, and addressing technical barriers. Consumer awareness and industry partnerships are also essential to build trust and stimulate demand for olive-based bio-products. Finally, strengthening supply chains and promoting collaboration among farmers, oil producers, and biorefineries will improve operational efficiency and ensure reliable feedstock availability.

In conclusion, the success of olive biomass biorefineries hinges on integrating innovative technologies, supportive policies, and market development. Holistic assessments must underpin these efforts to ensure scalability, economic viability, and sustainability. A SWOT analysis (Fig. 9) synthesizes the strengths, weaknesses, opportunities, and threats associated with olive biomass biorefineries, offering a structured

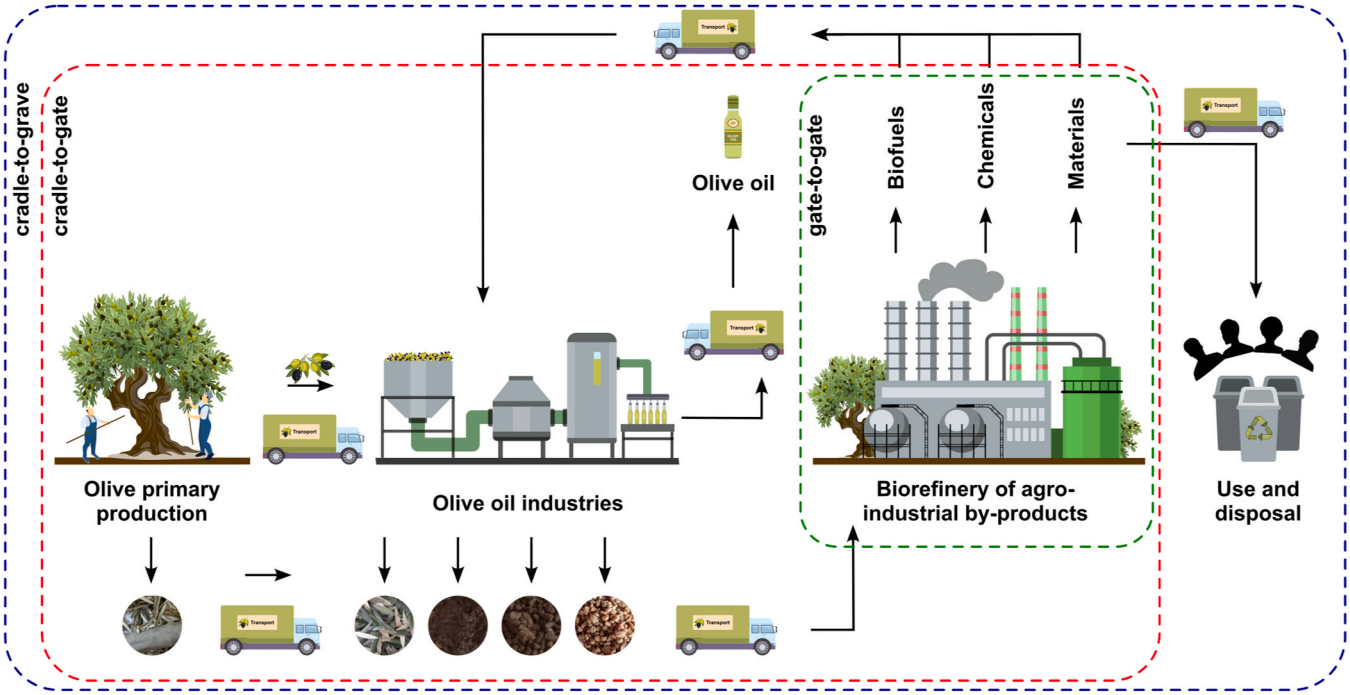


Fig. 8. Pictorial representation of gate-to-gate, cradle-to-gate, and cradle-to-grave approaches for life cycle analysis on olive biomass.

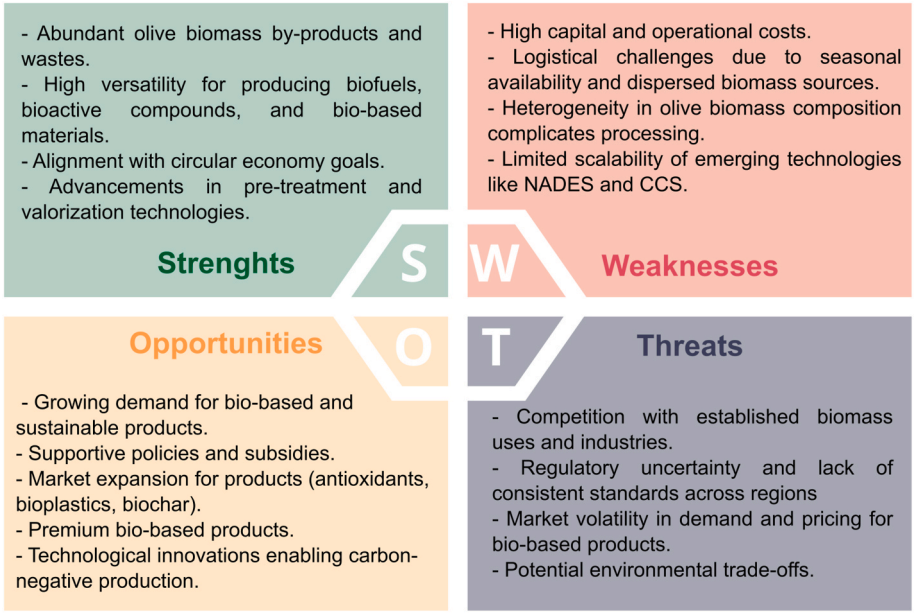


Fig. 9. SWOT analysis highlighting the strengths, weaknesses, opportunities, and threats associated with olive-based biorefineries. CCS: Carbon capture and storage; NADES: Natural Deep Eutectic Solvents.

perspective on their future potential. In this context, while governments frequently advocate for the circular bioeconomy, in practice, there remains limited direct financial support for businesses involved in waste valorization [159]. Nonetheless, there are initiatives in the Mediterranean region to valorize olive biomass, with most strategies focusing on bioenergy/energy products (heat, electricity, and biodiesel), olive pomace oil, or fertilizers. Other proposals include producing bioenergy, cosmetics, animal feed ingredients, and construction materials [158]. Efforts are also focused on gasification and pyrolysis to produce syngas and biochar [160,161].

9. Conclusions

Olive biomass biorefineries offer significant potential for sustainable production, leveraging diverse feedstocks such as olive tree pruning, leaves, pomace, and pits to produce biofuels, biochemicals, and bio-based materials. This review provides a comprehensive and multidimensional perspective on recent studies in olive biomass valorization in the biorefinery framework. Essential progress includes the development of innovative transformation pathways and pretreatment technologies to produce high-value products and renewable energy such as biofuels (ethanol, methane, and hydrogen), biochemicals (phenolic compounds, succinic acid, furfural, etc.), and bio-based materials (activated carbons, carbon dots, cellulose fibers, nanoparticles, biopolymers, etc.).

Despite these advances, challenges remain for large-scale implementation of olive biomass biorefineries, including scalability, cost reduction, and addressing environmental trade-offs. Future research should prioritize process optimization, integration of sustainable technologies, and the development of flexible, adaptive systems. By addressing these challenges, olive biomass biorefineries can play a transformative role in fostering the bioeconomy and achieving considerable environmental and economic benefits.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Chat-GPT in the writing process to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

[1] Solarte-Toro JC, Cardona Alzate CA. Biorefineries as the base for accomplishing the sustainable development goals (SDGs) and the transition to bioeconomy: technical aspects, challenges and perspectives. Bioresour Technol 2021;340:125626. https://doi.org/10.1016/j.biortech.2021.125626.
[2] Contreras MdM, Romero I, Moya M, Castro E. Olive-derived biomass as a renewable source of value-added products. Process Biochem 2020;97:43–56. https://doi.org/10.1016/j.procbio.2020.06.013.
[3] Zafeiroudis A, Kamperidou V, Barboutis I. Utilization of olive tree pruning residues in wood pellets. Eur J Wood Wood Prod 2024;82:1713–24. https://doi.org/10.1007/s00107-024-02114-x.
[4] Gómez-Cruz I, Contreras MdM, Romero I, Castro E. Towards the integral valorization of olive pomace-derived biomasses through biorefinery strategies. ChemBioEng Rev 2024;11:253–77. https://doi.org/10.1002/cben.202300045.
[5] Gómez-Cruz I, Contreras MdM, Romero I, Castro E. A biorefinery approach to obtain antioxidants, lignin and sugars from exhausted olive pomace. J Ind Eng Chem 2021;96:356–63. https://doi.org/10.1016/j.jiec.2021.01.042.
[6] Carrillo-Beltran R, Corpas-Iglesias FA, Terrones-Saeta JM, Bertoya-Sol M. New geopolymers from industrial by-products: olive biomass fly ash and chamotte as raw materials. Constr Build Mater 2021;272:121924. https://doi.org/10.1016/j.conbuildmat.2020.121924.

[7] Gul E, Al Bkoor Alrawashdeh K, Masek O, Skreiberg Ø, Corona A, Zampilli M, et al. Production and use of biochar from lignin and lignin-rich residues (such as digestate and olive stones) for wastewater treatment. J Anal Appl Pyrolysis 2021; 158:105263. https://doi.org/10.1016/j.jaap.2021.105263.
[8] Pérez-Almada D, Galán-Martín Á, Contreras MdM, Castro E. Integrated techno-economic and environmental assessment of biorefineries: review and future research directions. Sustain Energy Fuels 2023;7:4031–50. https://doi.org/10.1039/D3SE00405H.
[9] Bernini R, Campo M, Cassiani C, Fochetti A, Ieri F, Lombardi A, et al. Polyphenol-rich extracts from agroindustrial waste and byproducts: results and perspectives according to the green chemistry and circular economy. J Agric Food Chem 2024; 72:12871–95. https://doi.org/10.1021/acs.jafc.4c00945.
[10] García Martín JF, Cuevas M, Feng C-H, Álvarez Mateos P, Torres García M, Sánchez S. Energetic valorisation of olive biomass: olive-tree pruning, olive stones and pomaces. Processes 2020;8:511. https://doi.org/10.3390/pr8050511.
[11] Lenzuni M, Converti A, Casazza AA. From laboratory- to industrial-scale plants: future of anaerobic digestion of olive mill solid wastes. Bioresour Technol 2024; 394:130317. https://doi.org/10.1016/j.biortech.2024.130317.
[12] Lo Giudice V, Faraone I, Bruno MR, Ponticelli M, Labanca F, Bisaccia D, et al. Olive trees By-Products as sources of bioactive and other industrially useful compounds: a systematic review. Molecules 2021;26. https://doi.org/10.3390/molecules26165081.
[13] Lustosa Filho JF, da Silva APF, Costa ST, Gomes HT, de Figueiredo T, Hernández Z. Biochars derived from olive mill byproducts: typology, characterization, and eco-efficient application in agriculture—A systematic review. Sustainability 2024;16. https://doi.org/10.3390/su16125004.
[14] Romero-García JM, Niño L, Martínez-Patiño C, Álvarez C, Castro E, Negro MJ. Biorefinery based on olive biomass. State of the art and future trends. Bioresour Technol 2014;159:421–32. https://doi.org/10.1016/j.biortech.2014.03.062.
[15] Ahmed PM, Fernández PM, de Figueroa LIC, Pajot HF. Exploitation alternatives of olive mill wastewater: production of value-added compounds useful for industry and agriculture. Biofuel Res J 2019;6:980–94. https://doi.org/10.18331/BRJ2019.6.2.4.
[16] Sayın B, Kaban G. Biotechnological innovations unleashing the potential of olive mill wastewater in added-value bioproducts. Foods 2024;13. https://doi.org/10.3390/foods13142245.
[17] Maggiotto G, Colangelo G, Milanese M, de Risi A. Thermochemical technologies for the optimization of olive wood biomass energy exploitation: a review. Energies 2023;16. https://doi.org/10.3390/en16196772.
[18] García-Maraver A, Zamorano M, Ramos-Ridao A, Díaz LF. Analysis of olive grove residual biomass potential for electric and thermal energy generation in Andalusia (spain). Renew Sustain Energy Rev 2012;16:745–51. https://doi.org/10.1016/j.rser.2011.08.040.
[19] de Dios Maeso G. Biomass valorization in the olive oil sector through pyrolysis and biochar technology: a comprehensive review of Spanish case. Master Thesis. Linnaeus University; 2024.
[20] Contreras MdM, Galán-Martín Á, Seixas N, da Costa Lopes AM, Silvestre A, Castro E. Deep eutectic solvents for improved biomass pretreatment: current status and future prospective towards sustainable processes. Bioresour Technol 2023;369:128396. https://doi.org/10.1016/j.biortech.2022.128396.
[21] Tapia-Quirós P, Granados M, Sentellas S, Saurina J. Microwave-assisted extraction with natural deep eutectic solvents for polyphenol recovery from agrifood waste: mature for scaling-up? Sci Total Environ 2024;912:168716. https://doi.org/10.1016/j.scitotenv.2023.168716.
[22] Cascajares M, Alcayde A, Salmerón-Manzano E, Manzano-Agugliaro F. The bibliometric literature on scopus and WoS: the medicine and environmental sciences categories as case of study. Int J Environ Res Public Health 2021;18. https://doi.org/10.3390/ijerph18115851.
[23] Van Eck NJ, Waltman L. VOSviewer, a computer program for bibliometric mapping. Scientometrics 2010;84:523–38.
[24] Ghanbari R, Anwar F, Alkharfy KM, Gilani A-H, Saari N. Valuable nutrients and functional bioactives in different parts of olive (Olea europaea L.)—A review. Int J Mol Sci 2012;13:3291–340. https://doi.org/10.3390/ijms13033291.
[25] MAPA. Anuario 2023. https://www.mapa.gob.es/es/estadistica/temas/publicaciones/anuario-de-estadistica/2023/default.aspx. [Accessed 20 November 2024].
[26] International Olive Council. Unidad de Asuntos Económicos y Promoción 2024. https://www.internationaloliveoil.org/que-hacemos/unidad-de-asuntos-econonicos-y-promocion/?lang=es#figures. [Accessed 3 November 2024].
[27] Contreras MdM, Gómez-Cruz I, Romero I, Castro E. Olive pomace-derived biomasses fractionation through a two-step extraction based on the use of ultrasounds: Chemical characteristics. Foods 2021;10:111. https://doi.org/10.3390/foods10010111.
[28] Ferreira AF. Biorefinery concept. In: Rabaçal M, Ferreira AF, Silva CAM, Costa M, editors. Biorefineries: targeting energy, high value products and waste valorisation. Cham: Springer International Publishing; 2017. p. 1–20. https://doi.org/10.1007/978-3-319-48288-0_1.
[29] Martín-Lara MA, Hernáinz F, Calero M, Blázquez G, Tenorio G. Surface chemistry evaluation of some solid wastes from olive-oil industry used for lead removal from aqueous solutions. Biochem Eng J 2009;44:151–9. https://doi.org/10.1016/j.bej.2008.11.012.
[30] Cequier E, Aguilera J, Balcells M, Canela-Garayoa R. Extraction and characterization of lignin from olive pomace: a comparison study among ionic liquid, sulfuric acid, and alkaline treatments. Biomass Convers Biorefin 2019;9: 241–52. https://doi.org/10.1007/s13399-019-00400-w.
[31] Doménech P, Duque A, Higueras I, Fernández JL, Manzanares P. Analytical characterization of water-soluble constituents in olive-derived By-Products. Foods 2021;10:1299. https://doi.org/10.3390/foods10061299.
[32] Padilla-Rascón C, Ruiz E, Romero I, Castro E, Oliva JM, Ballesteros I, et al. Valorisation of olive stone by-product for sugar production using a sequential acid/steam explosion pretreatment. Ind Crops Prod 2020;148:112279. https://doi.org/10.1016/j.indcrop.2020.112279.
[33] Romero-García JM, López-Linares JC, Contreras MdM, Romero I, Castro E. Exploitation of olive tree pruning biomass through hydrothermal pretreatments. Ind Crops Prod 2022;176:114425. https://doi.org/10.1016/j.indcrop.2021.114425.
[34] Gómez-Cruz I, Contreras MdM, Romero I, Castro E. Sequential extraction of hydroxytyrosol, mannitol and triterpenic acids using a green optimized procedure based on ultrasound. Antioxidants 2021;10. https://doi.org/10.3390/antiox10111781.
[35] Gómez-Cruz I, Contreras MdM, Romero I, Castro E. Optimization of microwave-assisted water extraction to obtain high value-added compounds from exhausted olive pomace in a biorefinery context. Foods 2022;11. https://doi.org/10.3390/foods11142002.
[36] Caballero AS, Romero-García JM, Castro E, Cardona CA. Supercritical fluid extraction for enhancing polyphenolic compounds production from olive waste extracts. J Chem Technol Biotechnol 2020;95:356–62. https://doi.org/10.1002/jctb.5907.
[37] García-Vargas MC, Contreras MDM, Castro E. Avocado-derived biomass as a source of bioenergy and bioproducts. Applied Sciences (Switzerland) 2020;10: 1–29. https://doi.org/10.3390/app10228195.
[38] Semaan J-N, Belandria V, Missaoui A, Sarh B, Gökalp I, Bostyn S. Energy analysis of olive pomace valorization via hydrothermal carbonization. Biomass Bioenergy 2022;165:106590. https://doi.org/10.1016/j.biombioe.2022.106590.
[39] Glushkov D, Nyashina G, Shvets A, Pereira A, Ramanathan A. Current status of the pyrolysis and gasification mechanism of biomass. Energies 2021;14. https://doi.org/10.3390/en14227541.
[40] Quintana-Gómez L, Ladero M, Calvo L. Enzymatic production of biodiesel from alperujo oil in supercritical CO2. J Supercrit Fluids 2021;171:105184. https://doi.org/10.1016/j.supflu.2021.105184.
[41] Padilla-Rascón C, Carvalheiro F, Duarte LC, Roseiro LB, Ruiz E, Castro E. An integrated olive stone biorefinery based on a two-step fractionation strategy. Ind Crops Prod 2022;187:115157. https://doi.org/10.1016/j.indcrop.2022.115157.
[42] Algarra M, Orfãos L dos, Alves CS, Moreno-Tost R, Pino-González MS, Jiménez-Jiménez J, et al. Sustainable production of carbon nanoparticles from olive pit biomass: understanding proton transfer in the excited state on carbon dots. ACS Sustain Chem Eng 2019;7:10493–500. https://doi.org/10.1021/acssuschemeng.9b00969.
[43] Massadeh M, Fandi K. Acetone-Butanol-Ethanol (ABE) production by anaerobic microflora growing on olive mill wastewater. J Biobased Mater Bioenergy 2014;8: 94–9. https://doi.org/10.1166/jbmb.2014.1402.
[44] Romero-García JM, Susmozas A, Padilla-Rascón C, Manzanares P, Castro E, Oliva JM, et al. Ethanol production from olive stones using different process strategies. Renew Energy 2022;194:1174–83. https://doi.org/10.1016/j.renene.2022.06.022.
[45] Oreoluwa Jokodola E, Narisetty V, Castro E, Durgapal S, Coulon F, Sindhu R, et al. Process optimisation for production and recovery of succinic acid using xylose-rich hydrolysates by Actinobacillus succinogenes. Bioresour Technol 2022; 344:126224. https://doi.org/10.1016/j.biortech.2021.126224.
[46] Papachristopoulos E, Tsiaras E, Papadakis VG, Coutelieris FA. Design of a biogas power plant that uses olive tree pruning and olive kernels in achaia, Western Greece. Sustainability 2024;16. https://doi.org/10.3390/su16010187.
[47] Bušić A, Marđetko N, Kundas S, Morzak G, Belskaya H, Ivančić Šantek M, et al. Bioethanol production from renewable raw materials and its separation and purification: a review. Food Technol Biotechnol 2018;56:289–311. https://doi.org/10.17113/ftb.56.03.18.5546.
[48] Puentes JG, Mateo S, Sánchez S, Roberto IC, Moya AJ. Bioconversion study of olive tree biomass hemicellulosic hydrolysates by Candida guilliermondii at different scales for ethanol and xylitol production. Fermentation 2023;9. https://doi.org/10.3390/fermentation9060553.
[49] Arminda M, Josúe C, Cristina D, Fabiana S, Yolanda M. Use of activated carbons for detoxification of a lignocellulosic hydrolysate: statistical optimisation. J Environ Manag 2021;296:113320. https://doi.org/10.1016/j.jenvman.2021.113320.
[50] Mamaní A, Maturano Y, Herrero L, Montoro L, Sardella F. Increase in fermentable sugars of olive tree pruning biomass for bioethanol production: application of an experimental design for optimization of alkaline pretreatment. Period Polytech - Chem Eng 2022;66:269–78. https://doi.org/10.3311/PPch.18247.
[51] Abu Tayeh HN, Azaizeh H, Gerchman Y. Circular economy in olive oil production – olive mill solid waste to ethanol and heavy metal sorbent using microwave pretreatment. Waste Manag 2020;113:321–8. https://doi.org/10.1016/j.wasman.2020.06.017.
[52] López-Linares JC, Romero-García JM, Romero I, Ruiz E, Castro E. Development of a biorefinery from olive mill leaves: comparison of different process configurations. Ind Crops Prod 2023;200:116813. https://doi.org/10.1016/j.indcrop.2023.116813.
[53] Oliva JM, Negro MJ, Álvarez C, Manzanares P, Moreno AD. Fermentation strategies for the efficient use of olive tree pruning biomass from a flexible biorefinery approach. Fuel 2020;277:118171. https://doi.org/10.1016/j.fuel.2020.118171.

[54] Rivera EC, Assumpção DC, Kwon HJ, Okonkwo CC, Ezeji TC, Filho RM, et al. Mechanistic modeling of redox balance effects on the fermentation of eucalyptus wood-derived xylose to acetone-butanol-ethanol. Biochem Eng J 2023;190:108738. https://doi.org/10.1016/j.bej.2022.108738.
[55] Waehner RS, Mendez BA, Giulietti AM. Olive black water as raw material for butanol production. Biol Waste 1988;23:215–20. https://doi.org/10.1016/0269-7483(88)90035-3.
[56] González-Peñas H, Lu-Chau TA, Eibes G, Lema JM. Energy requirements and economics of acetone–butanol–ethanol (ABE) extractive fermentation: a solvent-based comparative assessment. Bioproc Biosyst Eng 2020;43:2269–81. https://doi.org/10.1007/s00449-020-02412-7.
[57] Aouine M, Elalami D, Koraichi SI, Haggoud A, Barakat A. Exploring natural fermented foods as a source for new efficient thermotolerant yeasts for the production of second-generation bioethanol. Energies 2022;15. https://doi.org/10.3390/en15144954.
[58] Peinado S, Mateo S, Sánchez S, Moya AJ. Effectiveness of sodium borohydride treatment on acid hydrolyzates from olive-tree pruning biomass for bioethanol production. Bioenergy Res 2019;12:302–11. https://doi.org/10.1007/s12155-019-09979-4.
[59] Ridha K, Dalila H, Ryma M, Mounir L, Abderrazaq H. Performance of a beer brewing enzyme in lignocellulosic biomass hydrolysis for bioethanol production. Biomass Convers Biorefin 2023. https://doi.org/10.1007/s13399-023-03990-8.
[60] Antonopoulou G, Kampranis A, Ntaikou I, Lyberatos G. Enhancement of liquid and gaseous biofuels production from agro-industrial residues after thermochemical and enzymatic pretreatment. Front Sustain Food Syst 2019;3. https://doi.org/10.3389/fsufs.2019.00092.
[61] Ryckebosch E, Drouillon M, Vervaeren H. Techniques for transformation of biogas to biomethane. Biomass Bioenergy 2011;35:1633–45. https://doi.org/10.1016/j.biombioe.2011.02.033.
[62] Astals S, Batstone DJ, Mata-Alvarez J, Jensen PD. Identification of synergistic impacts during anaerobic co-digestion of organic wastes. Bioresour Technol 2014;169:421–7. https://doi.org/10.1016/j.biortech.2014.07.024.
[63] Ochando-Pulido JM, Vuppala S, García-López AI, Martínez-Férez A. A focus on anaerobic digestion and co-digestion strategies for energy recovery and digestate valorization from olive-oil mill solid and liquid by-products. Sep Purif Technol 2024;333:125827. https://doi.org/10.1016/j.seppur.2023.125827.
[64] Hernández D, Pinilla F, Rebolledo-Leiva R, Aburto-Hole J, Díaz J, Quijano G, et al. Anaerobic Co-Digestion of agro-industrial waste mixtures for biogas production: an energetically sustainable solution. Sustainability 2024;16. https://doi.org/10.3390/su16062565.
[65] Valenti F, Porto SMC, Selvaggi R, Pecorino B. Co-digestion of by-products and agricultural residues: a bioeconomy perspective for a Mediterranean feedstock mixture. Sci Total Environ 2020;700:134440. https://doi.org/10.1016/j.scitotenv.2019.134440.
[66] Valenti F, Selvaggi R, Pecorino B, Porto SMC. Bioeconomy for sustainable development of biomethane sector: potential and challenges for agro-industrial by-products. Renew Energy 2023;215:119014. https://doi.org/10.1016/j.renene.2023.119014.
[67] Aravani VP, Tsigkou K, Papadakis VG, Kornaros M. Biochemical Methane potential of most promising agricultural residues in northern and southern Greece. Chemosphere 2022;296:133985. https://doi.org/10.1016/j.chemosphere.2022.133985.
[68] Fernández-Rodríguez MJ, de la Lama-Calvente D, García-González M, Moreno-Fernández J, Jiménez-Rodríguez A, Borja R, et al. Integral valorization of two-phase olive mill solid waste (OMSW) and related washing waters by anaerobic Co-digestion of OMSW and the microalga Raphidocelis subcapitata cultivated in these effluents. J Agric Food Chem 2022;70:3219–27. https://doi.org/10.1021/acs.jafc.1c08100.
[69] Fernández-Rodríguez MJ, de la Lama-Calvente D, Jiménez-Rodríguez A, Borja R, Rincón-Llorente B. Influence of the cell wall of Chlamydomonas reinhardtii on anaerobic digestion yield and on its anaerobic co-digestion with a carbon-rich substrate. Process Saf Environ Prot 2019;128:167–75. https://doi.org/10.1016/j.psep.2019.05.041.
[70] de la Lama-Calvente D, Fernández-Rodríguez MJ, Llanos J, Mancilla-Leytón JM, Borja R. Enhancing methane production from the invasive macroalga Rugulopteryx okamurae through anaerobic co-digestion with olive mill solid waste: process performance and kinetic analysis. J Appl Phycol 2021;33:4113–24. https://doi.org/10.1007/s10811-021-02548-3.
[71] Elalami D, Carrere H, Abdelouahdi K, Garcia-Bernet D, Peydecastaing J, Vaca-Medina G, et al. Mild microwaves, ultrasonic and alkaline pretreatments for improving methane production: impact on biochemical and structural properties of olive pomace. Bioresour Technol 2020;299:122591. https://doi.org/10.1016/j.biortech.2019.122591.
[72] Cubero-Cardoso J, Fernández-Prior Á, Ramiro-Garcia J, Trujillo-Reyes A, Caballero-Guerrero B, Rodríguez-Gutiérrez G, et al. Microbial population dynamics during unstable operation of a semicontinuous anaerobic digester fed with a mild-treated olive mill solid waste. Processes 2023;11. https://doi.org/10.3390/pr11092724.
[73] Tamborrino A, Catalano F, Leone A, Bianchi B. A real case study of a full-scale anaerobic digestion plant powered by olive By-Products. Foods 2021;10. https://doi.org/10.3390/foods10081946.
[74] Sözer S. A research on biogas production from a mixture of olive pomace and cattle manure. Biomass Convers Biorefin 2024;14:10651–9. https://doi.org/10.1007/s13399-023-04405-4.
[75] Serrano A, Fermoso FG, Alonso-Fariñas B, Rodríguez-Gutiérrez G, López S, Fernandez-Bolaños J, et al. Performance evaluation of mesophilic semi-continuous anaerobic digestion of high-temperature thermally pre-treated olive mill solid waste. Waste Manag 2019;87:250–7. https://doi.org/10.1016/j.wasman.2019.02.003.
[76] Almeida PV, Rodrigues RP, Slezak R, Quina MJ. Effect of phenolic compound recovery from agro-industrial residues on the performance of pyrolysis process. Biomass Convers Biorefin 2022;12:4257–69. https://doi.org/10.1007/s13399-021-02292-1.
[77] Girardi F, Mascitti A, d'Alessandro N, Tonucci L, Marzo GA, Remetti R. Effects of oxidative treatments on biomethane potential of solid olive residues. Waste Biomass Valorization 2023;14:1525–38. https://doi.org/10.1007/s12649-022-01987-4.
[78] Fernández-Prior Á, Trujillo-Reyes Á, Serrano A, Rodríguez-Gutiérrez G, Reinhard C, Fermoso FG. Biogas potential of the side streams obtained in a novel phenolic extraction system from olive mill solid waste. Molecules 2020;25:5438. https://doi.org/10.3390/molecules25225438.
[79] Yildirim O, Ozkaya B. Enhancing fermentation yield for biohydrogen production using eco-friendly nickel and cobalt ferrite nanoparticles. Biomass Convers Biorefin 2024. https://doi.org/10.1007/s13399-024-05354-2.
[80] Yildirim O, Tunay D, Ozkaya B, Demir A. Optimization of oxalic and sulphuric acid pretreatment conditions to produce bio-hydrogen from olive tree biomass. Int J Hydrogen Energy 2022;47:26316–25. https://doi.org/10.1016/j.ijhydene.2021.11.017.
[81] Carmo-Calado L, Hermoso-Orzáez MJ, Diaz-Perete D, Cal-Herrera J La, Brito P, Terrados-Cepeda J. Experimental research on the production of hydrogen-rich synthesis gas via the air-gasification of olive pomace: a comparison between an updraft bubbling bed and a downdraft fixed bed. Hydrogen 2023;4:726–45. https://doi.org/10.3390/hydrogen4040046.
[82] Smoliński A, Howaniec N. Experimental investigation and chemometric analysis of gasification and co-gasification of olive pomace and Sida hermaphrodita blends with sewage sludge to hydrogen-rich gas. Energy 2023;284:129208. https://doi.org/10.1016/j.energy.2023.129208.
[83] Tezer Ö, Karabağ N, Öngen A, Ayol A. Gasification performance of olive pomace in updraft and downdraft fixed bed reactors. Int J Hydrogen Energy 2023;48:22909–20. https://doi.org/10.1016/j.ijhydene.2023.02.088.
[84] Catizzone E, Freda C, Villone A, Romanelli A, Cornacchia G. Gasification of olive tree pruning in a rotary kiln reactor integrated with radio frequency plasma torch. Fuel 2025;381:133480. https://doi.org/10.1016/j.fuel.2024.133480.
[85] Dönmez F, Aker V, Ayas N. The evaluation of Ni–Co/Al2O3 via olive pomace pyrolysis to generate hydrogen-rich gas: experimental and kinetic study. Int J Hydrogen Energy 2022;47:19538–54. https://doi.org/10.1016/j.ijhydene.2022.02.031.
[86] Kypritidou Z, El-Bassi L, Jellali S, Kinigopoulou V, Tziritis E, Akrout H, et al. Lead removal from aqueous solutions by olive mill wastes derived biochar: batch experiments and geochemical modelling. J Environ Manag 2022;318:115562. https://doi.org/10.1016/j.jenvman.2022.115562.
[87] Castejón-del Pino R, Cayuela ML, Sánchez-García M, Sánchez-Monedero MA. Nitrogen availability in biochar-based fertilizers depending on activation treatment and nitrogen source. Waste Manag 2023;158:76–83. https://doi.org/10.1016/j.wasman.2023.01.007.
[88] Galán-Martín Á, Contreras MdM, Romero I, Ruiz E, Bueno-Rodríguez S, Eliche-Quesada D, et al. The potential role of olive groves to deliver carbon dioxide removal in a carbon-neutral Europe: opportunities and challenges. Renew Sustain Energy Rev 2022;165:112609. https://doi.org/10.1016/j.rser.2022.112609.
[89] Rodriguez-Fernandez S, Díaz I, González-Miquel M, González EJ, Rodriguez M. Optimization-based analysis of integrated lignocellulosic biorefineries in Spain focusing on building blocks. Biomass Convers Biorefin 2024;14:435–50. https://doi.org/10.1007/s13399-021-02247-6.
[90] Romero-García JM, Solarte-Toro JC, Galán-Martín Á, Ruiz E, Castro E, Ortiz-Sánchez M, et al. Olive leaves upgrading applying a novel two-stage organosolv pretreatment: techno-economic and environmental assessment. Biochem Eng J 2024;207:109317. https://doi.org/10.1016/j.bej.2024.109317.
[91] Servian-Rivas LD, Pachón ER, Rodríguez M, González-Miquel M, González EJ, Díaz I. Techno-economic and environmental impact assessment of an olive tree pruning waste multiproduct biorefinery. Food Bioprod Process 2022;134:95–108. https://doi.org/10.1016/j.fbp.2022.05.003.
[92] Osei JBD, Amiri A, Wang J, Tavares MT, Kiatkittipong W, Najdanovic-Visak V. Recovery of oils and antioxidants from olive stones. Biomass Bioenergy 2022;166:106623. https://doi.org/10.1016/j.biombioe.2022.106623.
[93] Loschi F, Zengin G, Zancato M, Ferrarese I, De Zordi N, Dall'Acqua S, et al. Valorisation of olive pomace from Veneto region: Oven-Dried material for food, nutraceutical, and cosmetic application of the OLIVARE project. Process Biochemistry 2024;137:153–63. https://doi.org/10.1016/j.procbio.2024.01.001.
[94] Katsinas N, Gehlsen U, García-Posadas L, Rodríguez-Rojo S, Steven P, González-García MJ, et al. Olive pomace phenolic compounds: from an agro-industrial By-Product to a promising ocular surface protection for dry eye disease. J Clin Med 2022;11. https://doi.org/10.3390/jcm11164703.
[95] Faraone I, Russo D, Bruno MR, Todaro L, D'Auria M, Milella L. Focus on L. pruning by-products: extraction techniques, biological activity, and phytochemical profile. Biofuels Bioprod Biorefining 2021;15:1835–49. https://doi.org/10.1002/bbb.2283.
[96] Conte P, Pulina S, Del Caro A, Fadda C, Urgeghe PP, De Bruno A, et al. Gluten-free breadsticks fortified with phenolic-rich extracts from olive leaves and olive mill wastewater. Foods 2021;10. https://doi.org/10.3390/foods10050923.
[97] Nicodemou A, Kallis M, Koutinas M. Biorefinery development for the production of polyphenols, algal biomass and lipids using olive processing industry waste.

Sustain Chem Pharm 2023;32:100998. https://doi.org/10.1016/j.scp.2023.100998.
[98] Ribeiro TB, Bonifácio-Lopes T, Morais P, Miranda A, Nunes J, Vicente AA, et al. Incorporation of olive pomace ingredients into yoghurts as a source of fibre and hydroxytyrosol: antioxidant activity and stability throughout gastrointestinal digestion. J Food Eng 2021;297:110476. https://doi.org/10.1016/j.jfoodeng.2021.110476.
[99] Paz A, Chalima A, Topakas E. Biorefinery of exhausted olive pomace through the production of polygalacturonases and omega-3 fatty acids by Crypthecodinium cohnii. Algal Res 2021;59:102470. https://doi.org/10.1016/j.algal.2021.102470.
[100] Freitas L, Simões R, Miranda I, Peres F, Ferreira-Dias S. Optimization of autohydrolysis of olive pomaces to obtain bioactive oligosaccharides: the effect of cultivar and fruit ripening. Catalysts 2022;12:788. https://doi.org/10.3390/catal12070788.
[101] Cox R, Narisetty V, Castro E, Agrawal D, Jacob S, Kumar G, et al. Fermentative valorisation of xylose-rich hemicellulosic hydrolysates from agricultural waste residues for lactic acid production under non-sterile conditions. Waste Manag 2023;166:336–45. https://doi.org/10.1016/j.wasman.2023.05.015.
[102] Romero-García JM, Fehér C, Cara C, Ruiz-Ramos E, Castro E. Improved Xylitol production from olive stones hydrolysates by biological detoxification. J Clean Prod 2022;336:130408. https://doi.org/10.1016/j.jclepro.2022.130408.
[103] Correia J, Gudiña EJ, Lazar Z, Janek T, Teixeira JA. Cost-effective rhamnolipid production by Burkholderia thailandensis E264 using agro-industrial residues. Appl Microbiol Biotechnol 2022;106:7477–89. https://doi.org/10.1007/s00253-022-12225-1.
[104] Cassoni AC, Costa P, Mota I, Vasconcelos MW, Pintado M. Recovery of lignins with antioxidant activity from Brewer's spent grain and olive tree pruning using deep eutectic solvents. Chem Eng Res Des 2023;192:34–43. https://doi.org/10.1016/j.cherd.2023.01.053.
[105] Tortosa G, Mesa S, Delgado MJ, Amaya-Gómez CV. "alperujo" compost improves nodulation and symbiotic nitrogen fixation of soybean inoculated with Bradyrhizobium diazoefficiens. Nitrogen 2023;4:223–30. https://doi.org/10.3390/nitrogen4020015.
[106] Óleo Revista. Falta de rentabilidad de las orujeras: bajo precio del aceite de orujo de oliva y del orujillo 2019. https://www.oleorevista.com/texto-diario/mostrar/3658171/falta-rentabilidad-orujeras-bajo-precio-aceite-orujo-oliva-orujillo. [Accessed 8 August 2024].
[107] Paz A, Karnaouri A, Templis CC, Papayannakos N, Topakas E. Valorization of exhausted olive pomace for the production of omega-3 fatty acids by Crypthecodinium cohnii. Waste Manag 2020;118:435–44. https://doi.org/10.1016/j.wasman.2020.09.011.
[108] de la Casa JA, Bueno JS, Castro E. Recycling of residues from the olive cultivation and olive oil production process for manufacturing of ceramic materials. A comprehensive review. J Clean Prod 2021;296:126436. https://doi.org/10.1016/j.jclepro.2021.126436.
[109] Pardalis N, Xanthopoulou E, Zamboulis A, Bikiaris DN. Olive stone as a filler for recycled high-density polyethylene: a promising valorization of solid wastes from olive oil industry. Sustainable Chemistry for the Environment 2024;6:100090. https://doi.org/10.1016/j.scenv.2024.100090.
[110] Martínez-Alvarenga H, Gutiérrez MC, Benítez A, Martín MA, Caballero A. Comprehensive study of valorisation of exhausted olive pomace through the preparation of highly porous activated carbons. Fuel 2024;374:132502. https://doi.org/10.1016/j.fuel.2024.132502.
[111] Schievano A, Adani F, Buessing L, Botto A, Casoliba EN, Rossoni M, et al. An integrated biorefinery concept for olive mill waste management: supercritical CO2 extraction and energy recovery. Green Chem 2015;17:2874–87. https://doi.org/10.1039/C5GC00076A.
[112] Sutay D, Yağcı S, Yurtdaş E, Toptaş M. Multiproduct biorefinery from defatted olive mill waste: preparation of hemicellulose-based biodegradable films and instant controlled pressure drop (DIC)-assisted isolation of value-added products. Biomass Convers Biorefin 2024;14:14627–39. https://doi.org/10.1007/s13399-023-03739-3.
[113] Fillat Ú, Wicklein B, Martín-Sampedro R, Ibarra D, Ruiz-Hitzky E, Valencia C, et al. Assessing cellulose nanofiber production from olive tree pruning residue. Carbohydr Polym 2018;179:252–61. https://doi.org/10.1016/j.carbpol.2017.09.072.
[114] Mabrouk A Ben, Putaux J-L, Boufi S. Valorization of olive leaf waste as a new source of fractions containing cellulose nanomaterials. Ind Crops Prod 2023;202:116996. https://doi.org/10.1016/j.indcrop.2023.116996.
[115] Gómez-Casero MA, Pérez-Villarejo L, Castro E, Eliche-Quesada D. Reinforcement of alkali-activated cements based matrices using olive pruning fibres as an alternative to traditional fibres. Sustain Chem Pharm 2024;37:101433. https://doi.org/10.1016/j.scp.2024.101433.
[116] Monteagudo JM, Durán A, Alonso M, Stoica A-I. Investigation of effectiveness of KOH-activated olive pomace biochar for efficient direct air capture of CO2. Sep Purif Technol 2025;352:127997. https://doi.org/10.1016/j.seppur.2024.127997.
[117] Vadequímica. Comprar carbón activado 2024. https://www.vadequimica.com/carbon-activo-25-k.html?c2c_material=polvo-180-8. [Accessed 7 December 2024].
[118] Cirillo C, Iuliano M, Scarpa D, Sarno M. Olive mill waste (OMW) as raw material for bio-lubricant production by an innovative immobilized lipase. Biocatal Agric Biotechnol 2024;57:103078. https://doi.org/10.1016/j.bcab.2024.103078.
[119] Alouiz I, Aqil M, Chari A, Dahbi M, Amarouch MY, Mazouzi D. Influence of phosphorus activation and carbonization temperature on the electrochemical performance of hard carbon made from olive pomace as an anode for sodium-ion batteries. RSC Adv 2025;15:19546–60. https://doi.org/10.1039/D5RA02547H.
[120] Coppola G, Gaudio MT, Lopresto CG, Calabro V, Curcio S, Chakraborty S. Bioplastic from renewable biomass: a facile solution for a greener environment. Earth Systems and Environment 2021;5:231–51. https://doi.org/10.1007/s41748-021-00208-7.
[121] Rodríguez-Liébana JA, Navas-Martos FJ, Jurado-Contreras S, Morillas-Gutiérrez F, Mateo S, Moya AJ, et al. Manufacture and characterisation of polylactic acid biocomposites with high-purity cellulose isolated from olive pruning waste. J Reinf Plast Compos 2024;43:318–38. https://doi.org/10.1177/07316844231162286.
[122] Bacha S, Arous F, Chouikh E, Jaouani A, Gtari M, Charradi K, et al. Exploring Bacillus amyloliquefaciens strain OM81 for the production of polyhydroxyalkanoate (PHA) bioplastic using olive mill wastewater. 3 Biotech 2023;13:415. https://doi.org/10.1007/s13205-023-03808-4.
[123] Acharjee SA, Bharali P, Gogoi B, Sorhie V, Walling B, Alemtoshi. PHA-based bioplastic: a potential alternative to address microplastic pollution. Water Air Soil Pollut 2022;234:21. https://doi.org/10.1007/s11270-022-06029-2.
[124] Nocito G, Sciuto EL, Franco D, Nastasi F, Pulvirenti L, Petralia S, et al. Physicochemical characterization and antibacterial properties of carbon dots from two mediterranean olive solid waste cultivars. Nanomaterials 2022;12. https://doi.org/10.3390/nano12050885.
[125] Rodríguez-Carballo G, Moreno-Tost R, Fernandes S, Esteves da Silva JCG, Pinto da Silva L, Galiano EC, et al. Nitrogen doped carbon dots as a photocatalyst based on biomass. A life cycle assessment. J Clean Prod 2023;423:138728. https://doi.org/10.1016/j.jclepro.2023.138728.
[126] Sousa DA, Ferreira LFV, Fedorov AA, Rego AMB do, Ferraria AM, Cruz AB, et al. Luminescent carbon dots from wet olive pomace: structural insights, photophysical properties and cytotoxicity. Molecules 2022;27. https://doi.org/10.3390/molecules27196768.
[127] Alcazar-Ruiz A, Maisano S, Chiodo V, Urbani F, Dorado F, Sanchez-Silva L. Enhancing CO2 capture performance through activation of olive pomace biochar: a comparative study of physical and chemical methods. Sustain Mater Technol 2024;42:e01177. https://doi.org/10.1016/j.susmat.2024.e01177.
[128] Dahdouh A, Khay I, Bouizi Y, Kervern G, Pontvianne S, El Maakoul A, et al. Hydrothermal carbonization of two-phase olive mill waste (alperujo): effect of aqueous phase recycling. Biomass Bioenergy 2024;184:107205. https://doi.org/10.1016/j.biombioe.2024.107205.
[129] Liuzzi S, Rubino C, Martellotta F, Stefanizzi P, Casavola C, Pappalettera G. Characterization of biomass-based materials for building applications: the case of straw and olive tree waste. Ind Crops Prod 2020;147:112229. https://doi.org/10.1016/j.indcrop.2020.112229.
[130] Pérez-Villarejo L, Eliche-Quesada D, Martín-Pascual J, Martín-Morales M, Zamorano M. Comparative study of the use of different biomass from olive grove in the manufacture of sustainable ceramic lightweight bricks. Constr Build Mater 2020;231:117103. https://doi.org/10.1016/j.conbuildmat.2019.117103.
[131] Mouzakitis Y, Adamides ED. Techno-economic assessment of an olive mill wastewater (OMWW) biorefinery in the context of circular bioeconomy. Eng 2022;3:488–503. https://doi.org/10.3390/eng3040035.
[132] Marquina J, Colinet MJ, Pablo-Romero M del P. The economic value of olive sector biomass for thermal and electrical uses in Andalusia (spain). Renew Sustain Energy Rev 2021;148:111278. https://doi.org/10.1016/j.rser.2021.111278.
[133] Lanfranchi M, Giannetto C, De Pascale A. Economic analysis and energy valorization of by-products of the olive oil process: "valdemone DOP" extra virgin olive oil. Renew Sustain Energy Rev 2016;57:1227–36. https://doi.org/10.1016/j.rser.2015.12.196.
[134] Campos P, Miller AZ, Knicker H, Costa-Pereira MF, Merino A, De la Rosa JM. Chemical, physical and morphological properties of biochars produced from agricultural residues: implications for their use as soil amendment. Waste Manag 2020;105:256–67. https://doi.org/10.1016/j.wasman.2020.02.013.
[135] Cardoza D, Romero I, Martínez T, Ruiz E, Gallego FJ, López-Linares JC, et al. Location of biorefineries based on olive-derived biomass in Andalusia, Spain. Energies 2021;14. https://doi.org/10.3390/en14113052.
[136] Zurob E, Cabezas R, Villarroel E, Rosas N, Merlet G, Quijada-Maldonado E, et al. Design of natural deep eutectic solvents for the ultrasound-assisted extraction of hydroxytyrosol from olive leaves supported by COSMO-RS. Sep Purif Technol 2020;248:117054. https://doi.org/10.1016/j.seppur.2020.117054.
[137] Plaza A, Tapia X, Yañez C, Vilches F, Candia O, Cabezas R, et al. Obtaining hydroxytyrosol from olive mill waste using deep eutectic solvents and then supercritical CO2. Waste Biomass Valorization 2020;11:6273–84. https://doi.org/10.1007/s12649-019-00836-1.
[138] Ünlü AE. Green and non-conventional extraction of bioactive compounds from olive leaves: screening of novel natural deep eutectic solvents and investigation of process parameters. Waste Biomass Valorization 2021;12:5329–46. https://doi.org/10.1007/s12649-021-01411-3.
[139] Pontes PV de A, Czaikoski A, Almeida NA, Fraga S, Rocha L de O, Cunha RL, et al. Extraction optimization, biological activities, and application in O/W emulsion of deep eutectic solvents-based phenolic extracts from olive pomace. Food Res Int 2022;161:111753. https://doi.org/10.1016/j.foodres.2022.111753.
[140] Carmona I, Aguirre I, Griffith DM, García-Borrego A. Towards a circular economy in virgin olive oil production: valorization of the olive mill waste (OMW) "alpeorujo" through polyphenol recovery with natural deep eutectic solvents (NADESs) and vermicomposting. Sci Total Environ 2023;872:162198. https://doi.org/10.1016/j.scitotenv.2023.162198.
[141] Romero-García JM, Sanchez A, Rendón-Acosta G, Martínez-Patiño JC, Ruiz E, Magaña G, et al. An olive tree pruning biorefinery for Co-Producing high value-added bioproducts and biofuels: economic and energy efficiency analysis. Bioenergy Res 2016;9:1070–86. https://doi.org/10.1007/s12155-016-9786-3.

[142] Susmozas A, Moreno AD, Romero-García JM, Manzanares P, Ballesteros M. Designing an olive tree pruning biorefinery for the production of bioethanol, xylitol and antioxidants: a techno-economic assessment. Holzforschung 2019;73:15–23. https://doi.org/10.1515/hf-2018-0099.
[143] Ramos JS, Ferreira AF. Techno-economic analysis and life cycle assessment of olive and wine industry co-products valorisation. Renew Sustain Energy Rev 2022;155:111929. https://doi.org/10.1016/j.rser.2021.111929.
[144] Manzanares P. The role of biorefinering research in the development of a modern bioeconomy. Acta Innovations 2020:47–56. https://doi.org/10.32933/actainnovations.37.4.
[145] Orive M, Cebrián M, Amayra J, Zufía J, Bald C. Integrated biorefinery process for olive pomace valorisation. Biomass Bioenergy 2021;149:106079. https://doi.org/10.1016/j.biombioe.2021.106079.
[146] Batuecas E, Tommasi T, Battista F, Negro V, Sonetti G, Viotti P, et al. Life cycle assessment of waste disposal from olive oil production: anaerobic digestion and conventional disposal on soil. J Environ Manag 2019;237:94–102. https://doi.org/10.1016/j.jenvman.2019.02.021.
[147] Khounani Z, Hosseinzadeh-Bandbafha H, Moustakas K, Talebi AF, Goli SAH, Rajaeifar MA, et al. Environmental life cycle assessment of different biorefinery platforms valorizing olive wastes to biofuel, phosphate salts, natural antioxidant, and an oxygenated fuel additive (triacetin). J Clean Prod 2021;278:123916. https://doi.org/10.1016/j.jclepro.2020.123916.
[148] Espadas-Aldana G, Guaygua-Amaguaña P, Vialle C, Belaud J-P, Evon P, Sablayrolles C. Life cycle assessment of olive pomace as a reinforcement in polypropylene and polyethylene biocomposite materials: a new perspective for the valorization of this agricultural By-Product. Coatings 2021;11. https://doi.org/10.3390/coatings11050525.
[149] Fernández-Lobato L, Aguado R, Jurado F, Vera D. Biomass gasification as a key technology to reduce the environmental impact of virgin olive oil production: a life cycle assessment approach. Biomass Bioenergy 2022;165:106585. https://doi.org/10.1016/j.biombioe.2022.106585.
[150] Fanourakis S, Romero-García JM, Castro E, Jiménez-Esteller L, Galán-Martín Á. Economic and environmental implications of carbon capture in an olive pruning tree biomass biorefinery. J Clean Prod 2024;456:142361. https://doi.org/10.1016/j.jclepro.2024.142361.
[151] Arias A, Feijoo G, Moreira MT. Biorefineries as a driver for sustainability: key aspects, actual development and future prospects. J Clean Prod 2023;418:137925. https://doi.org/10.1016/j.jclepro.2023.137925.
[152] Ioannou I, Galán-Martín Á, Pérez-Ramírez J, Guillén-Gosálbez G. Trade-offs between sustainable development goals in carbon capture and utilisation. Energy Environ Sci 2023;16:113–24. https://doi.org/10.1039/D2EE01153K.
[153] Ozturk MU, Ayol A, Tezer O. Life cycle assessment of olive pomace gasification for an up-draft fixed bed gasifier system. Int J Hydrogen Energy 2023;48:23339–47. https://doi.org/10.1016/j.ijhydene.2023.01.206.
[154] Patzschke CF, Bahzad H, Boot-Handford ME, Fennell PS. Simulation of a 100-MW solar-powered thermo-chemical air separation system combined with an oxy-fuel power plant for bio-energy with carbon capture and storage (BECCS). Mitig Adapt Strateg Glob Chang 2020;25:539–57. https://doi.org/10.1007/s11027-019-09879-0.
[155] Cadena E, Rocca F, Gutierrez JA, Carvalho A. Social life cycle assessment methodology for evaluating production process design: biorefinery case study. J Clean Prod 2019;238:117718. https://doi.org/10.1016/j.jclepro.2019.117718.
[156] Ruiz E, Romero-García JM, Romero I, Manzanares P, Negro MJ, Castro E. Olive-derived biomass as a source of energy and chemicals. Biofuel Bioprod Biorefining 2017;11:1077–94. https://doi.org/10.1002/bbb.1812.
[157] Asghar A, Sairash S, Hussain N, Baqar Z, Sumrin A, Bilal M. Current challenges of biomass refinery and prospects of emerging technologies for sustainable bioproducts and bioeconomy. Biofuel Bioprod Biorefining 2022;16:1478–94. https://doi.org/10.1002/bbb.2403.
[158] Sepahi N, Ilinca A, Rousse DR. Multi-criteria decision analysis for evaluating carbon capture technologies in power plants. Renew Sustain Energy Rev 2025;219:115699. https://doi.org/10.1016/j.rser.2025.115699.
[159] Donner M, Radić I, Erraach Y, El Hadad-Gauthier F. Implementation of circular business models for olive oil waste and By-Product valorization. Resources 2022;11. https://doi.org/10.3390/resources11070068.
[160] Olive pomace gasification - BioRural. n.d. https://biorural.eu/success-stories/olive-pomace-gasification/. [Accessed 7 July 2025].
[161] Carboliva: primera planta de biochar a partir de orujillo de aceituna | AVEBIOM n.d. https://www.avebiom.org/biomasanews/avebiom/carboliva-primera-planta-de-biochar-partir-de-orujillo-de-aceituna (accessed July 7, 2025).